



LABORATORY ACTIVITY NO. 2.3.8

Using Scripting Tools

Name: Garcia, Marina Ellise V.

Section: BSIT-102I

This lab begins by teaching the creation of a simple batch file to automate the generation of a network configuration report. It then moves to Visual Basic Scripting, demonstrating how to create a basic loop program and showcasing how VBScripts can be executed by double-clicking them or via the *cscript* command in the command prompt. The VBScript exercise progresses to include more complex features like generating random numbers and using conditional statements with message boxes. Finally, the lab introduces Windows PowerShell, illustrating how to use cmdlets to retrieve system information and filter/sort results. It also covers creating and saving PowerShell scripts in the PowerShell ISE, and addresses the security measure of execution policies, demonstrating how to allow unsigned scripts to run.

I. OBJECTIVES

Upon completion of this laboratory exercise, students are expected to and meet the following intended learning outcomes:

Activity Objectives	Intended Learning Outcomes		
	ILO1	ILO2	ILO3
1) create and execute basic batch scripts to automate command-line tasks, understanding file saving procedures and command prompt interactions.	•	•	•
2) Develop and test Visual Basic Scripts (VBScripts) to perform operations such as loops and conditional logic, and understand how to run these scripts from both File Explorer and the command.	•	•	
3) Utilize Windows PowerShell to run cmdlets for system information retrieval and manipulate output using pipes and formatting. Additionally, learn to create and execute PowerShell scripts, including managing the execution policy for security.	•	•	

Where:

ILO1 – Exhibit thorough demonstration of the concept

ILO2 – Being able to perform the activity independently



ILO3 – Design and evaluate emerging IT solutions

II. DISCUSSION

- **Programming Logic**

- Focuses on software development, programming languages, and the importance of clear program sequence, input, processing, and output.
- Illustrates program flow using flowcharts and explains the concept of pseudocode, emphasizing breaking down steps into simple instructions and using subroutines.

- **Features of Programming Code**

- Highlights common constructs across programming languages, including subroutines (user-defined), return statements, conditional statements (if, if-else), variables, functions (pre-defined), user interfaces (console, GUI, web), and comments.

- **Programming Languages**

- Discusses factors in choosing a language, such as program type, programmer experience, and target devices.
- Lists popular languages like Java, Python, JavaScript, C++, C#, PHP, Perl, and Ruby.
- Differentiates between **compiled languages** (transformed to executable binary, fast but platform-specific, e.g., C++, C#) and **interpreted languages** (run within an interpreter, slower but platform-independent, e.g., Java, JavaScript, Python).
- Covers **query languages** (for databases, e.g., SQL), **assembly language** (human-readable machine code), and **markup languages** (for document structure, e.g., HTML, XML).

- **Identifiers**

- Explains variables (values that can change) and constants (values that do not change) in programming.
- Introduces identifiers for multiple elements like arrays (fixed-size, same data type) and vectors (dynamic arrays).

- **Branches and Loops**



- Describes control flow mechanisms: branches (execute code based on condition, e.g., If...Then) and loops (repeat code based on condition, e.g., For...Next, Do While).
- **Operators**
 - Details comparison operators (e.g., ==, !=, <, >) and logical operators (AND, OR, XOR).
- **Procedures, Functions, and Comments**
 - Explains that procedures and functions are used for blocks of code, with functions specifically returning a value.
 - Highlights the use of comments for documenting code, noting they are not executed by the compiler/interpreter.
- **Object-Oriented Programming (OOP)**
 - Introduces the concept of designing code to represent "things" (objects) that interact through defined methods.
 - Defines attributes (values/data types of an object), methods (what can be done to an object), and properties (alternative public access to fields).
- **Scripting Languages**
 - Describes scripting languages as interpreted, often used for OS automation, smaller than programs, and designed for specific tasks (e.g., Batch files, Windows PowerShell, VBScript, Bash).
- **Application Platforms and Delivery Methods**
 - Categorizes applications by platform (single-platform, cross-platform, web applications) and delivery methods (local installation, network installation, cloud-hosted).

III. MATERIALS

A Windows 11 installed computer system.

IV. PROCEDURE

For this activity, you will execute the provided step-by-step instructions. As you complete each step, capture a screenshot of the resulting output or completed action that directly demonstrates successful execution of that instruction. Paste each screenshot into the designated area within your lab document.



Important Considerations:

- **Accessing Notepad (Windows 11):**
 - Instead of "Press START, type notepad and then press ENTER", in Windows 11, you would click the **Start button** (usually centered on the taskbar), type notepad in the search bar that appears, and then press Enter.
- **Accessing Command Prompt (Windows 11):**
 - To open Command Prompt, instead of "Click Start, scroll down and expand Windows System, and then click Command Prompt", in Windows 11, click the **Start button**, type cmd in the search bar, and then click "Command Prompt" from the search results.
- **Accessing Windows Explorer (Windows 11):**
 - Instead of "Open Windows Explorer and navigate to your Documents folder", in Windows 11, you can simply click the **File Explorer icon** on the taskbar (looks like a folder) or press Windows key + E, then navigate to your Documents folder.
- **Accessing Windows PowerShell / PowerShell ISE (Windows 11):**
 - For "Click Start, type Windows PowerShell, right-click Windows PowerShell and then click Run as administrator" , and "Click Start, type Windows PowerShell, right-click Windows PowerShell ISE and then click Run as administrator", in Windows 11:
 - Click the **Start button**.
 - Type Windows PowerShell or Windows PowerShell ISE in the search bar.
 - Right-click the relevant application in the search results and select "Run as administrator."
 - **User Account Control (UAC) Prompts:** Be prepared to click "Yes" on any User Account Control (UAC) prompts that appear when running applications or making system changes as an administrator.
 - **File Extensions:** Pay close attention to saving files with the correct extensions: .cmd for batch files , .vbs for Visual Basic Scripts , and .ps1 for PowerShell scripts. Incorrect extensions will prevent the scripts from running as intended.
 - **Script Content Accuracy:** Type the script code exactly as provided in the instructions, including special characters and spacing, as syntax errors can prevent scripts from running.



- **PowerShell Execution Policy:** Understand that PowerShell has a default execution policy that blocks unsigned scripts for security. The lab specifically instructs you to change this policy to

RemoteSigned to allow your created script to run. It's crucial to be aware of the security implications when modifying execution policies on a production system.

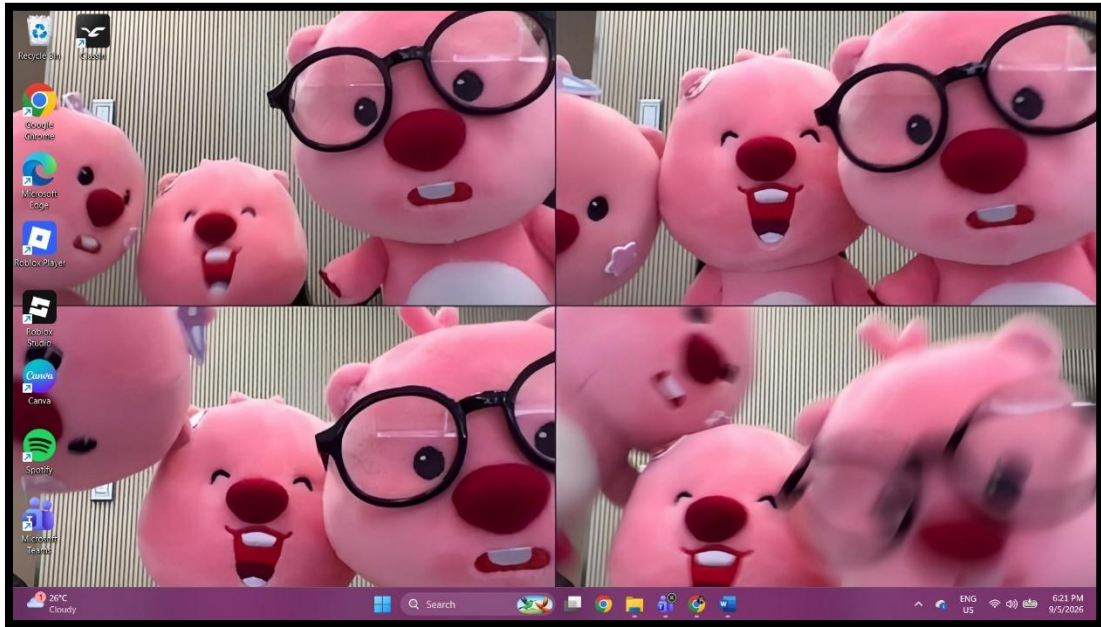
- **Piping Output:** The PowerShell exercises demonstrate the use of the pipe (|) operator to send the output of one command (cmdlet) as input to another, which is a fundamental concept in PowerShell for chaining commands.
- **Command Line vs. ISE:** The lab highlights the difference between running PowerShell cmdlets directly in the command line and creating and executing scripts within the PowerShell Integrated Scripting Environment (ISE), which offers features like color-coding and autocompletion to aid in development.



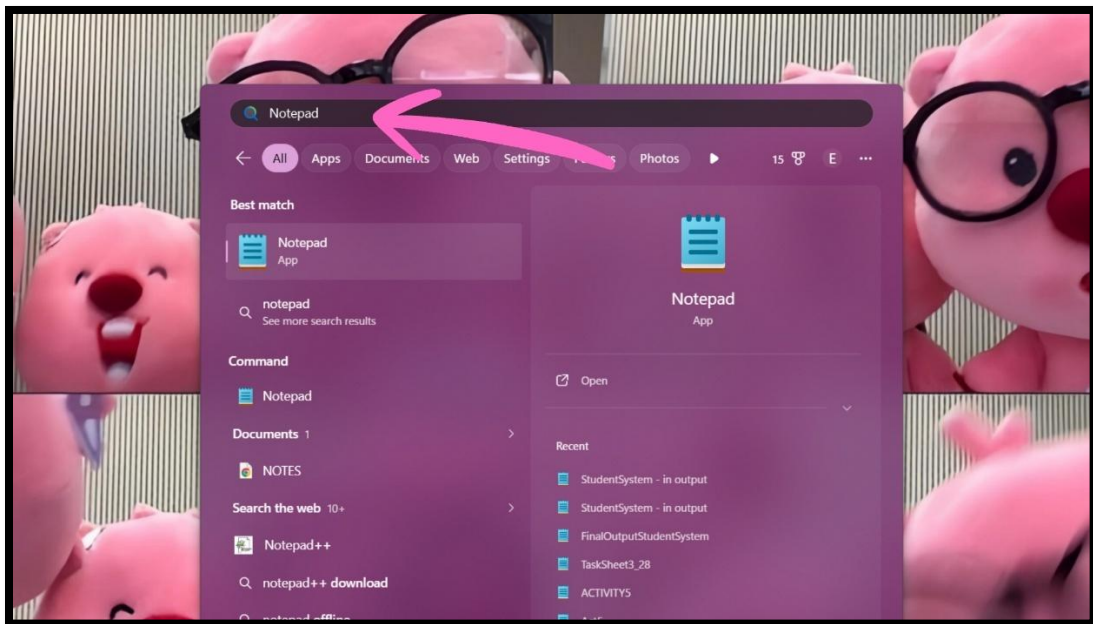
Steps:

Part 1: Creating a Batch File

1. Start your computer and sign in.



2. Click the **Start button** (on the taskbar), type notepad in the search bar, and then press ENTER.





3. In the Untitled - Notepad window, type the following text:

```
@echo off  
ipconfig /all > report.txt  
echo Press a key to view report  
pause > nul  
notepad report.txt
```

o **Note the features of the script:**

- **@echo off** - suspends output being written to the command prompt.
- **ipconfig /all > report.txt** - calls the system command to produce a network configuration summary (ipconfig) and redirects (or pipes) its output to a text file.
- The remainder of the script waits for the user to press a key and then opens the report file in Notepad.

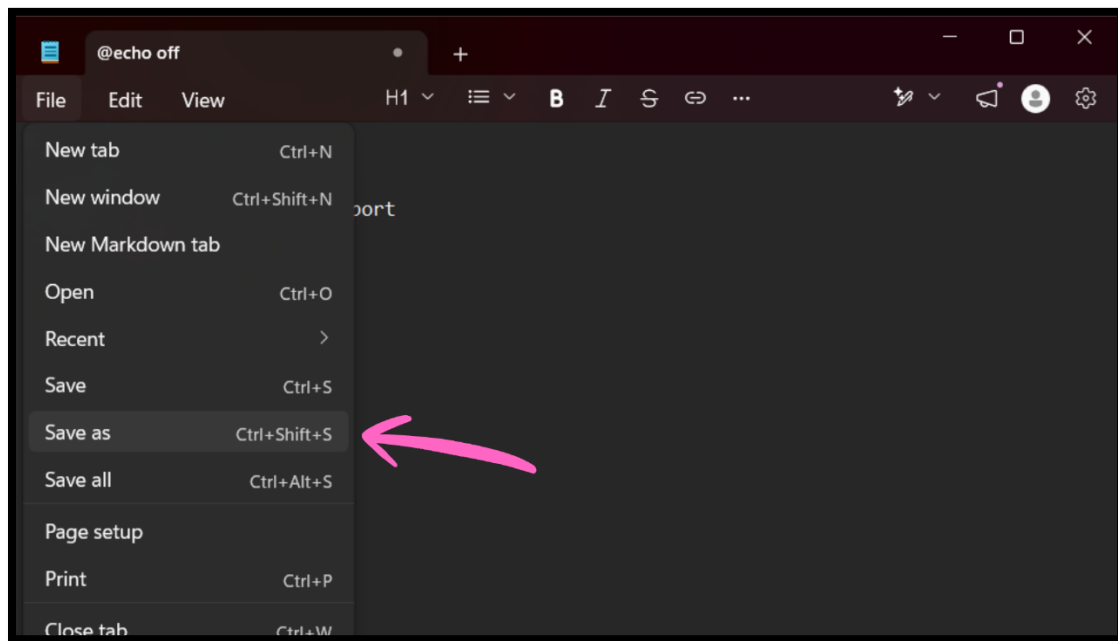
The screenshot shows a Notepad window with a dark theme. The title bar reads '@echo off'. The menu bar includes 'File', 'Edit', and 'View'. The toolbar shows various formatting options like 'H1', 'B', 'I', 'S', and 'G'. The text area contains the following script:

```
@echo off  
ipconfig /all > report.txt  
echo Press a key to view report  
pause > nul  
notepad report.txt
```

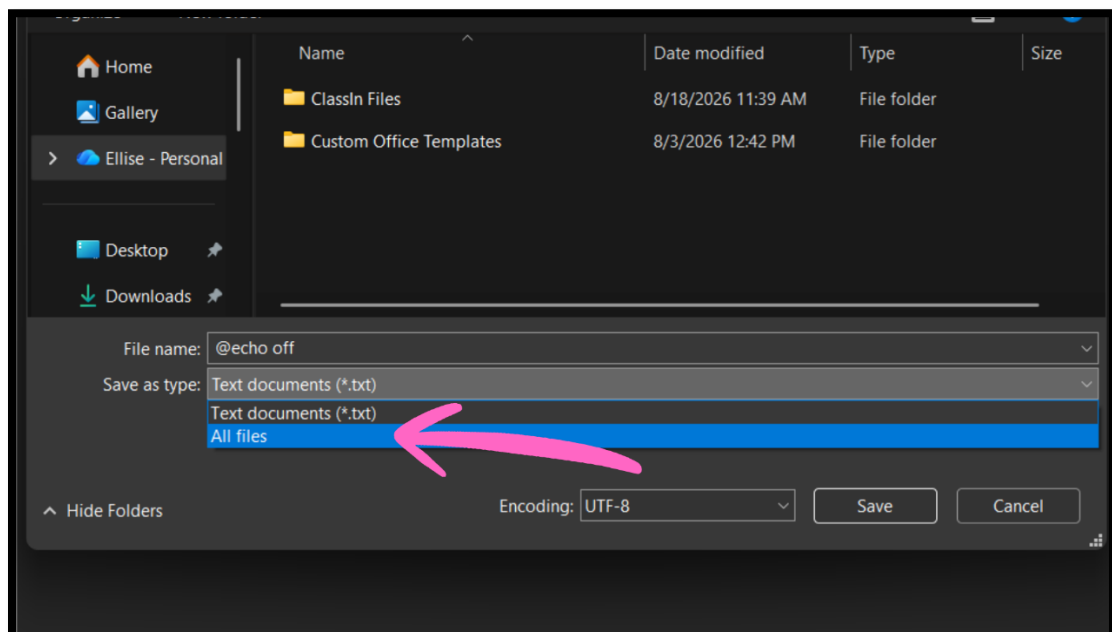
A pink arrow points to the 'pause > nul' line, highlighting it.



4. From the File menu, select **Save As**.

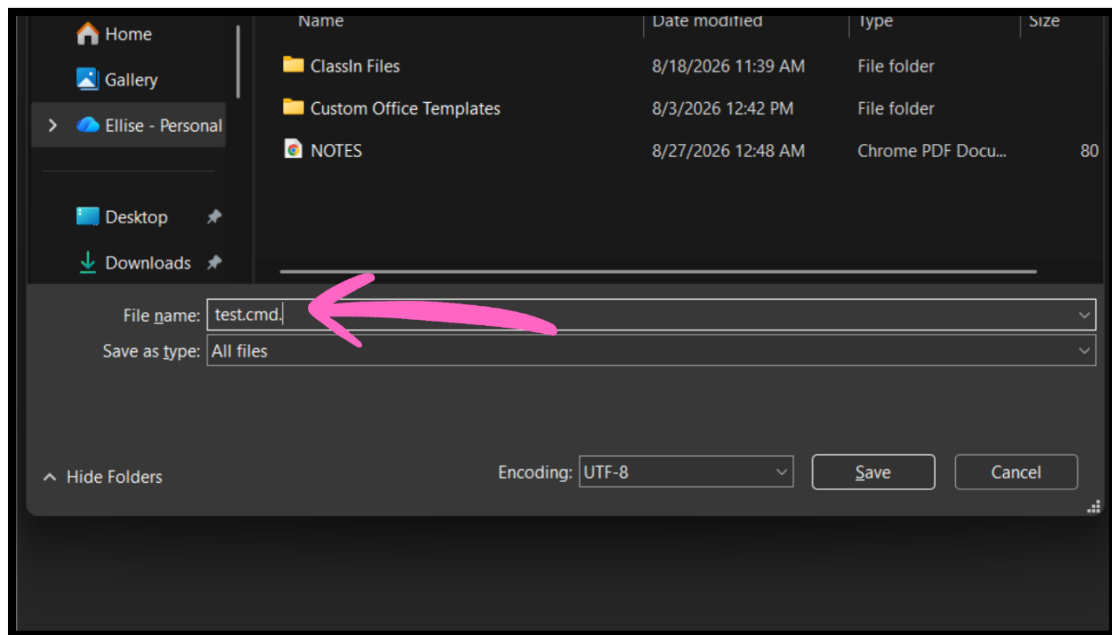


5. In the Save as type list, click **All Files**.

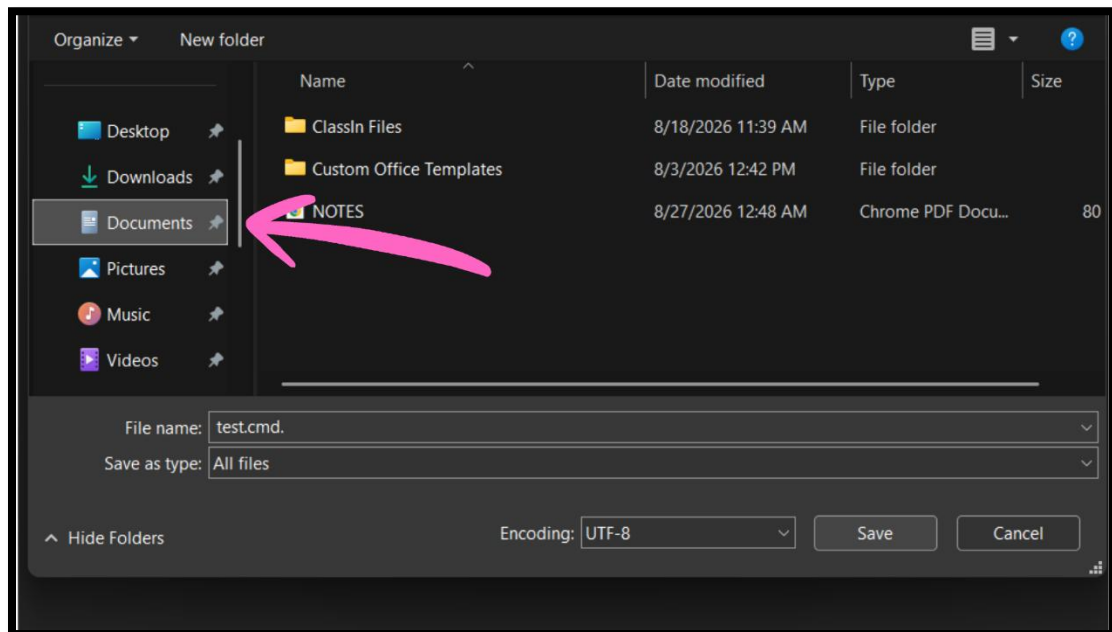




6. In the File name box, type test.cmd.

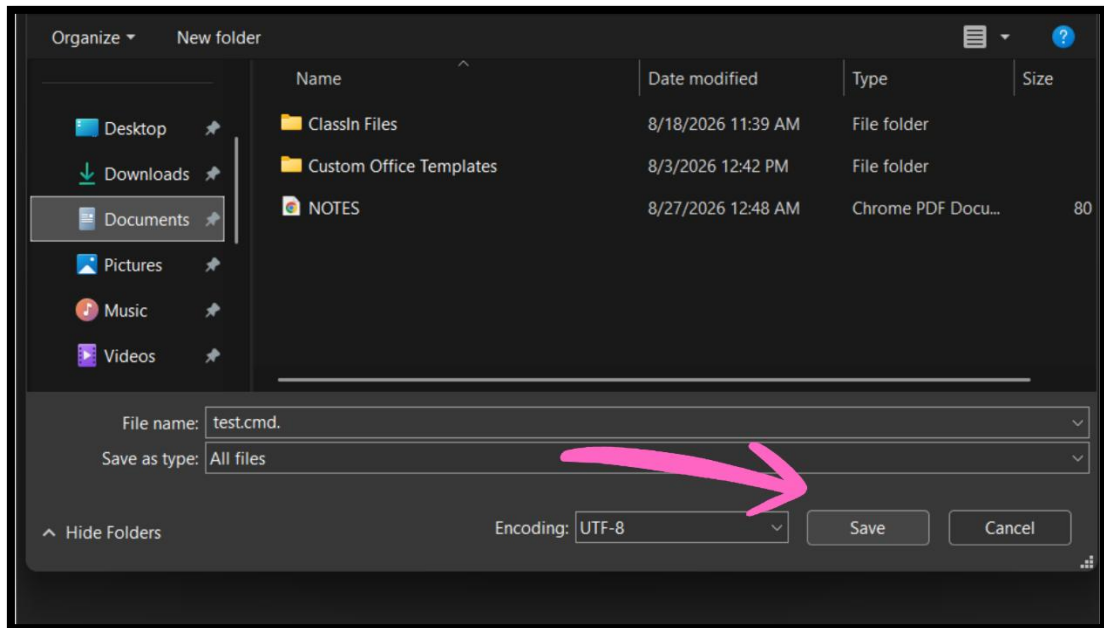


7. In the navigation pane, click **Documents**.

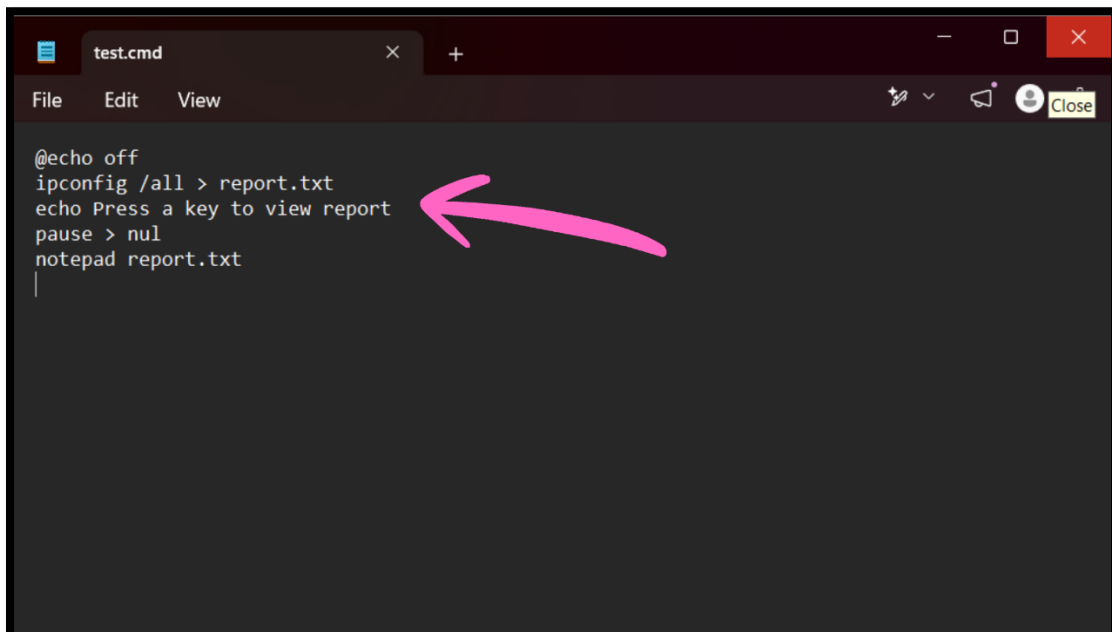




8. Click **Save**.

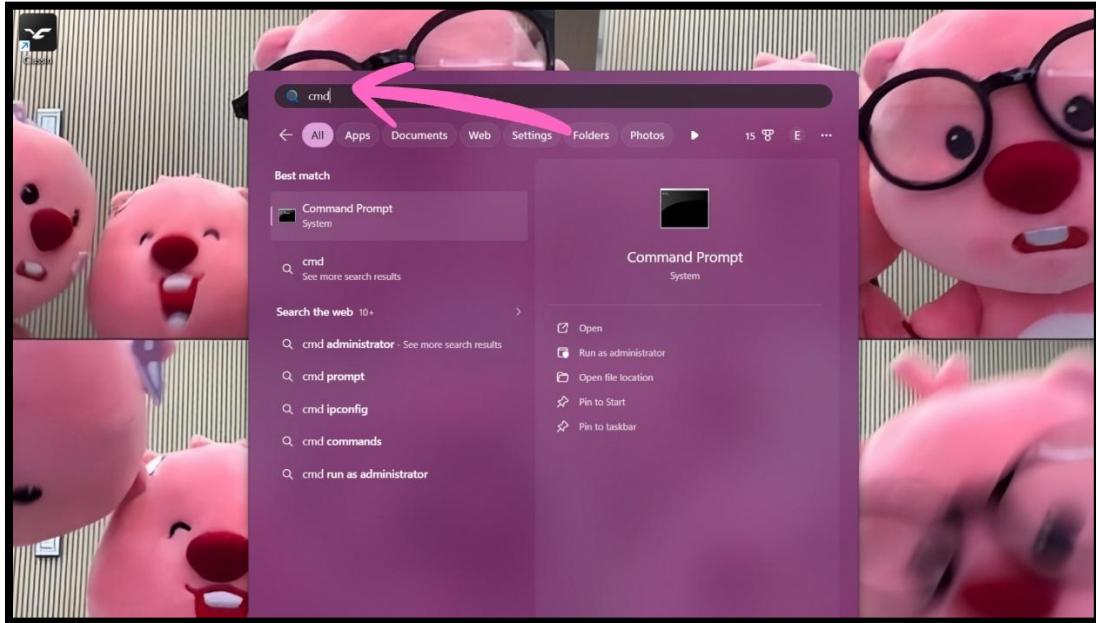


9. Close Notepad.

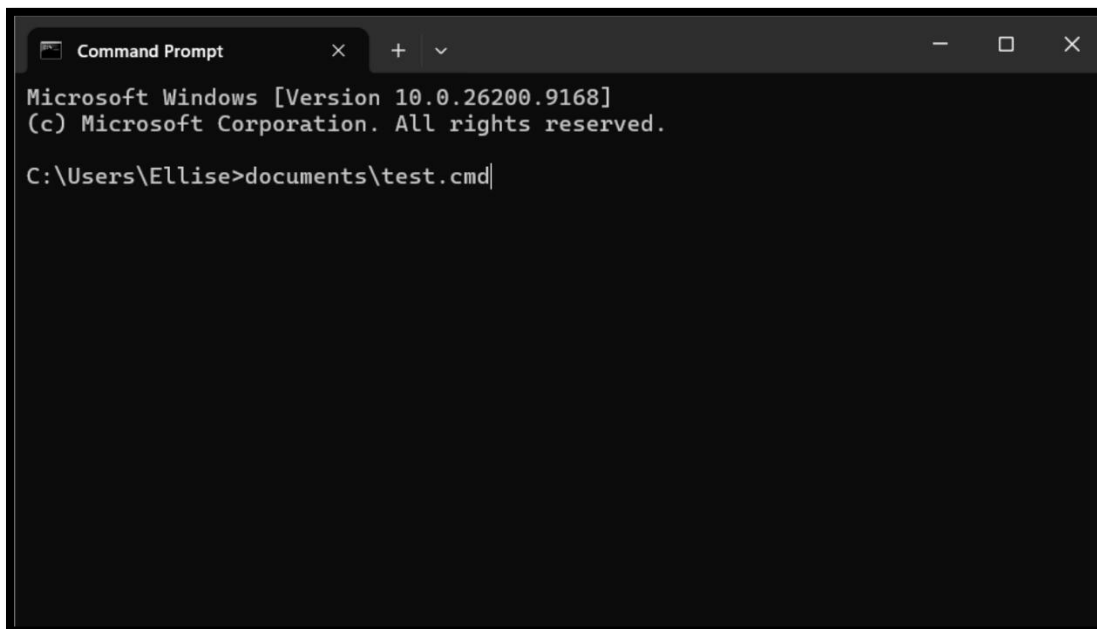




10. Click the **Start button**, type cmd in the search bar, and then click **Command Prompt** from the search results.



11. In the Command Prompt window, type documents\test.cmd and then press ENTER.





12. When prompted to press a key, press **SPACEBAR**.

- A Notepad window opens and displays the output from the ipconfig.exe command that you put in your batch file.

A screenshot of a Notepad window titled "report.txt" with a dark theme. The text inside shows the output of the Windows IP Configuration command. The window has a menu bar with "File", "Edit", and "View", and a toolbar with various icons. The text is as follows:

```
Windows IP Configuration

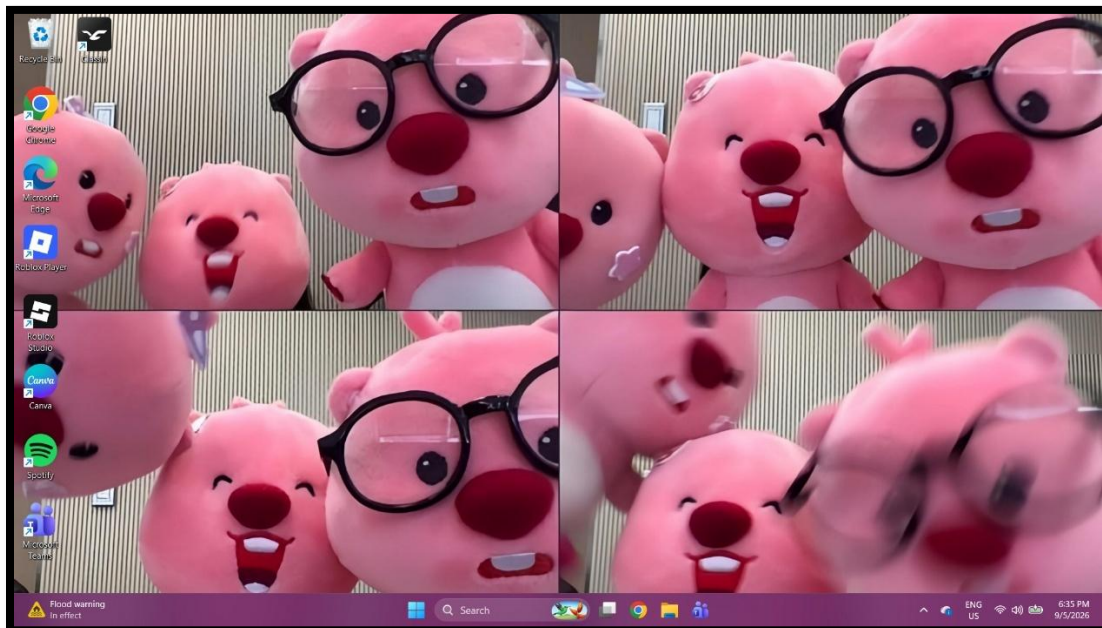
Host Name . . . . . : LAPTOP-T9ABQOHL
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . : E3-WDS12.COM
Description . . . . . : Realtek PCIe GbE Family Controller
Physical Address. . . . . : BC-FC-E7-8E-F3-E6
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 1:
```

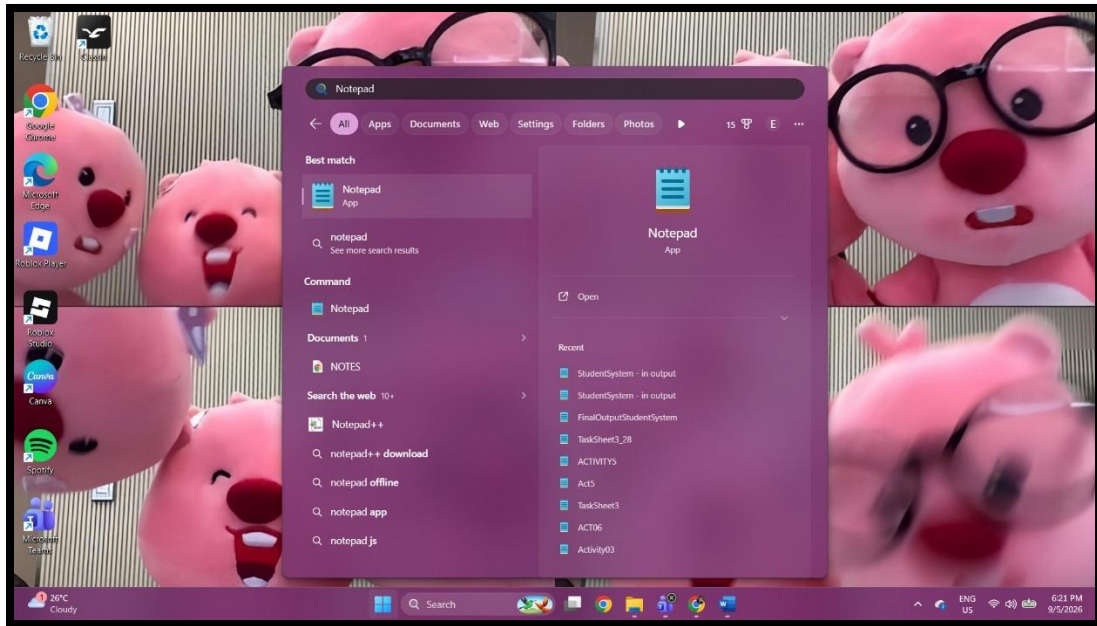
13. Close all open windows and apps.





Part 2: Creating a Visual Basic Script

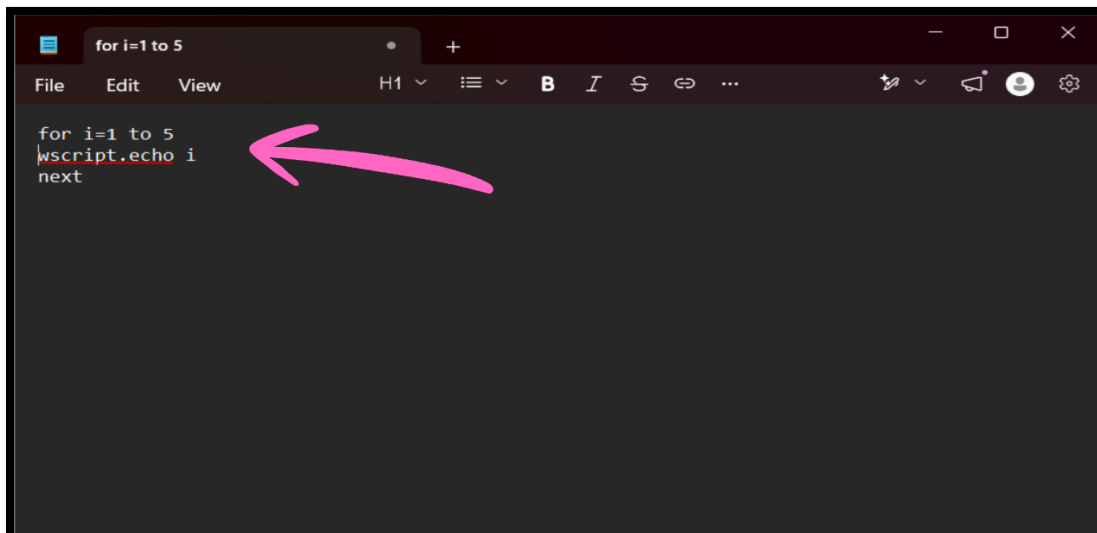
1. Click the **Start button**, type notepad in the search bar, and then press ENTER.



2. In the Untitled - Notepad window, type the following text:

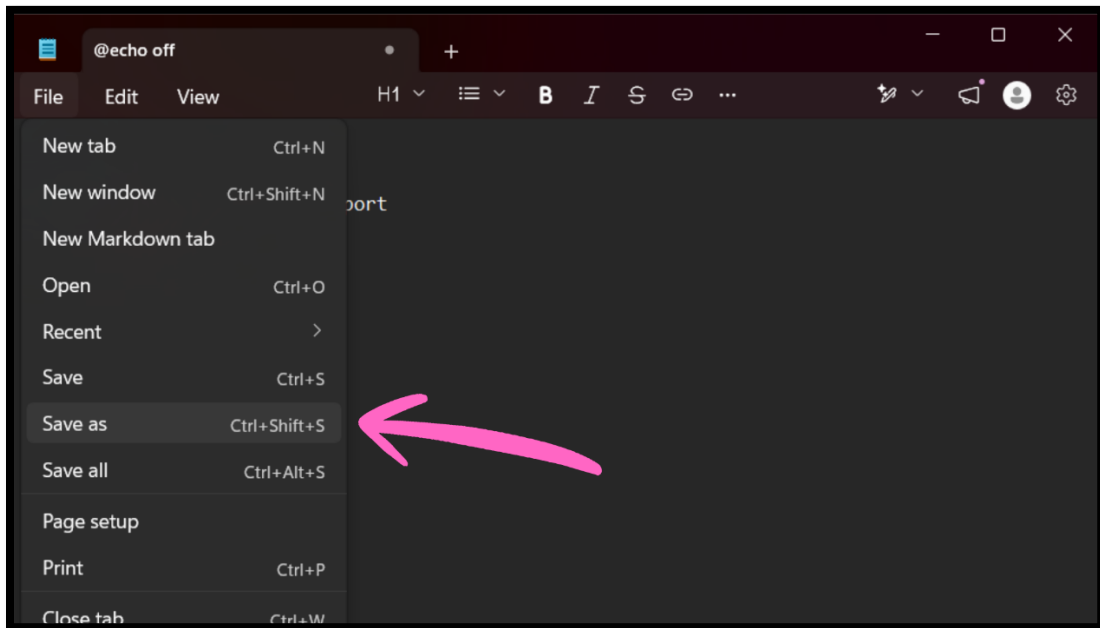
```
for i=1 to 5  
    wscript.echo i  
next
```

- This program creates a simple loop and uses the echo statement to show the current value of the variable *i* as the loop executes.

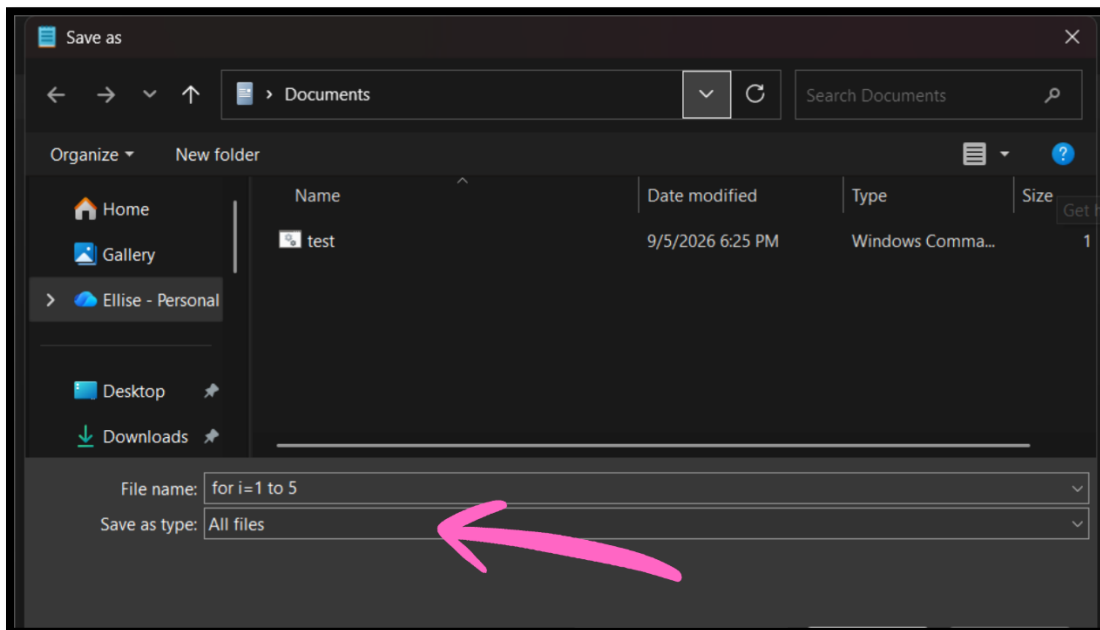




3. From the File menu, select **Save As**.

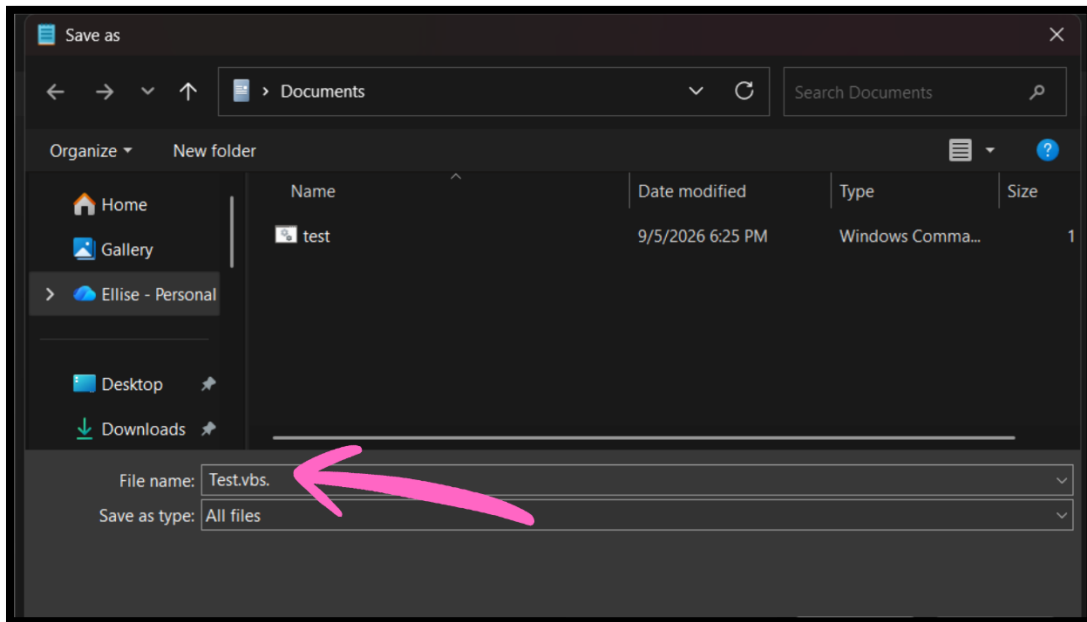


4. In the Save as type list, click **All Files**.

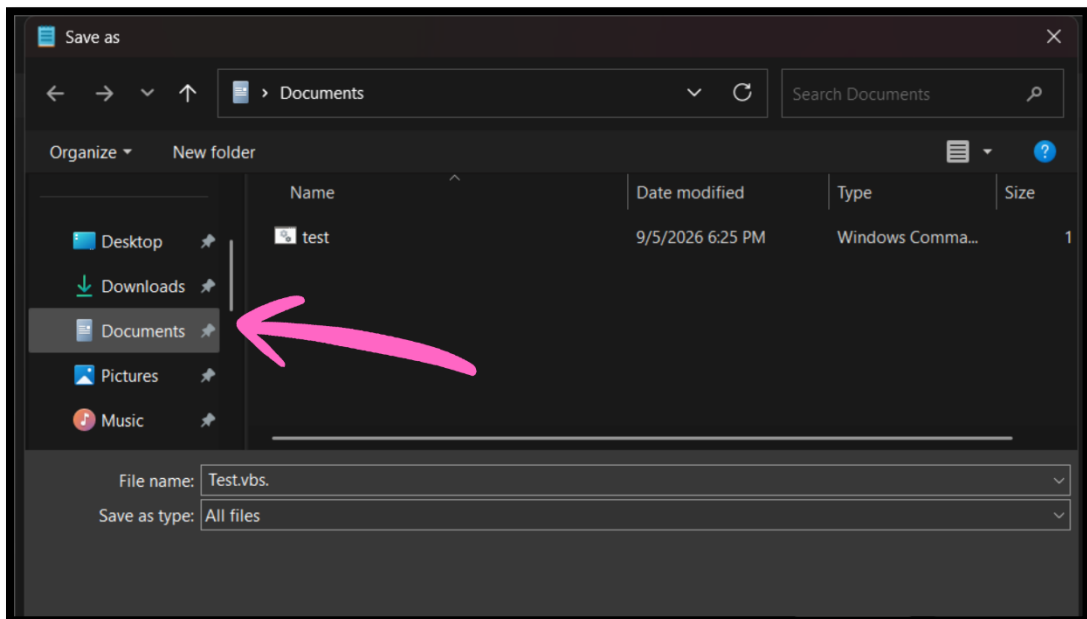




5. In the File name box, type *Test.vbs*.

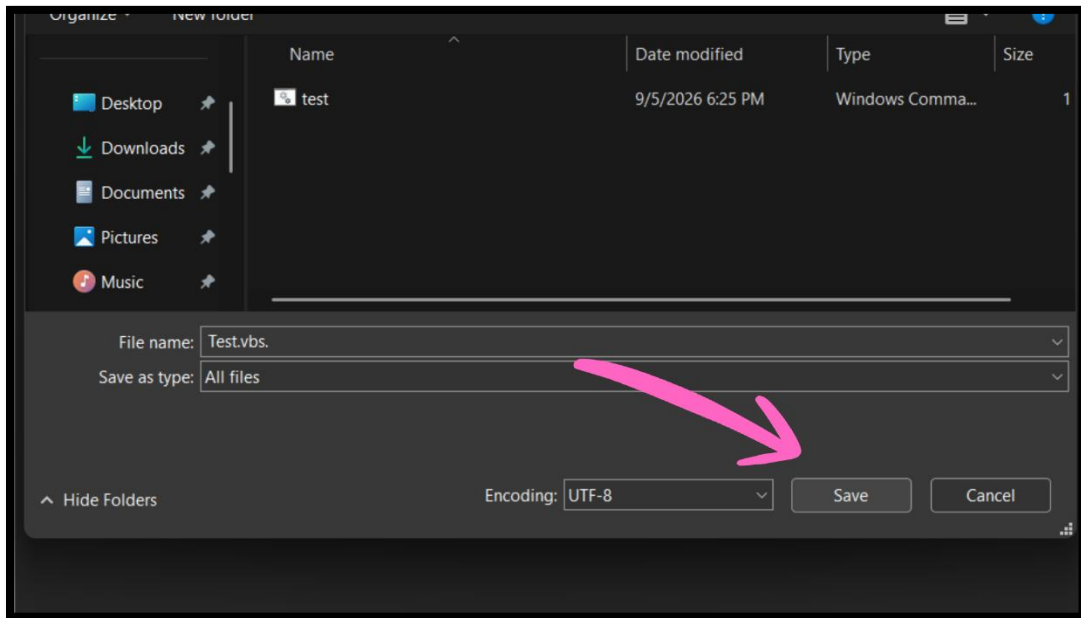


6. In the navigation pane, click **Documents**.

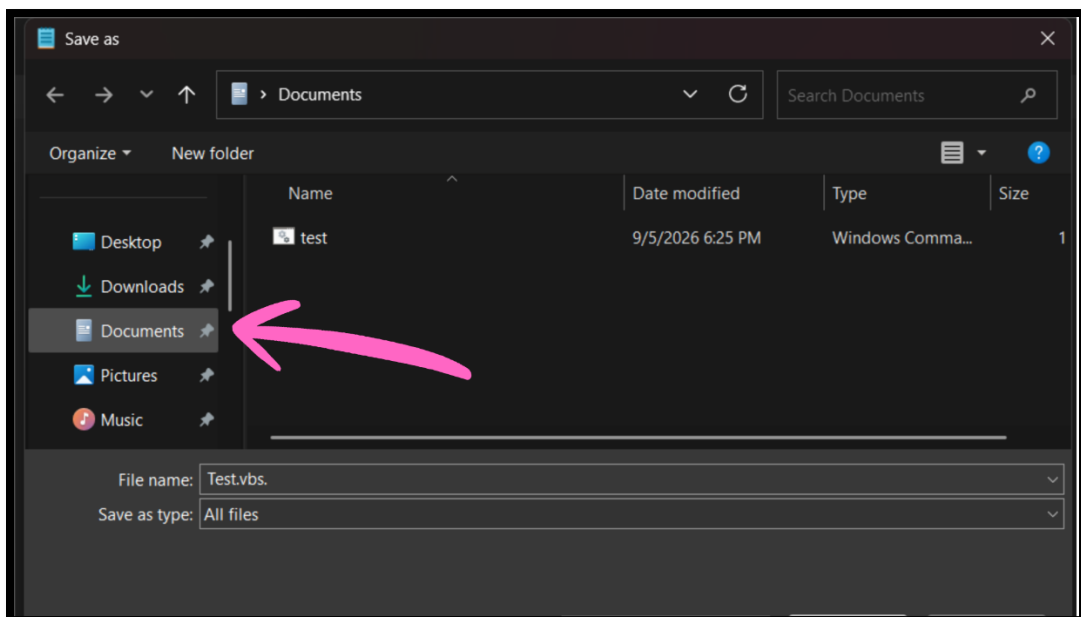




7. Click **Save**.

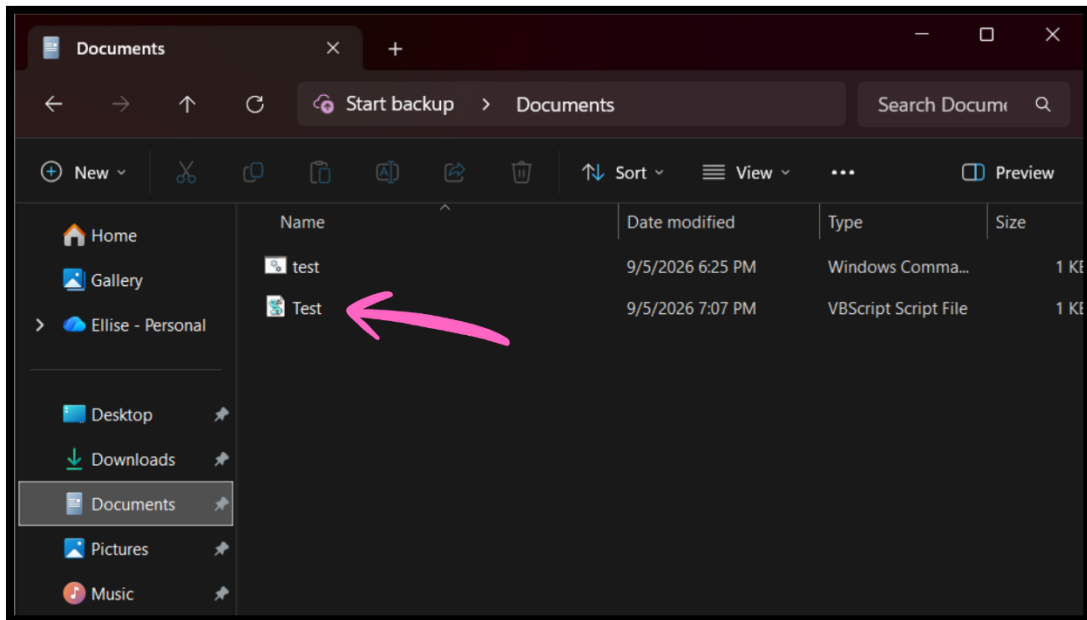


8. Open **File Explorer** (by clicking the folder icon on the taskbar or pressing Windows key + E) and navigate to your Documents folder.

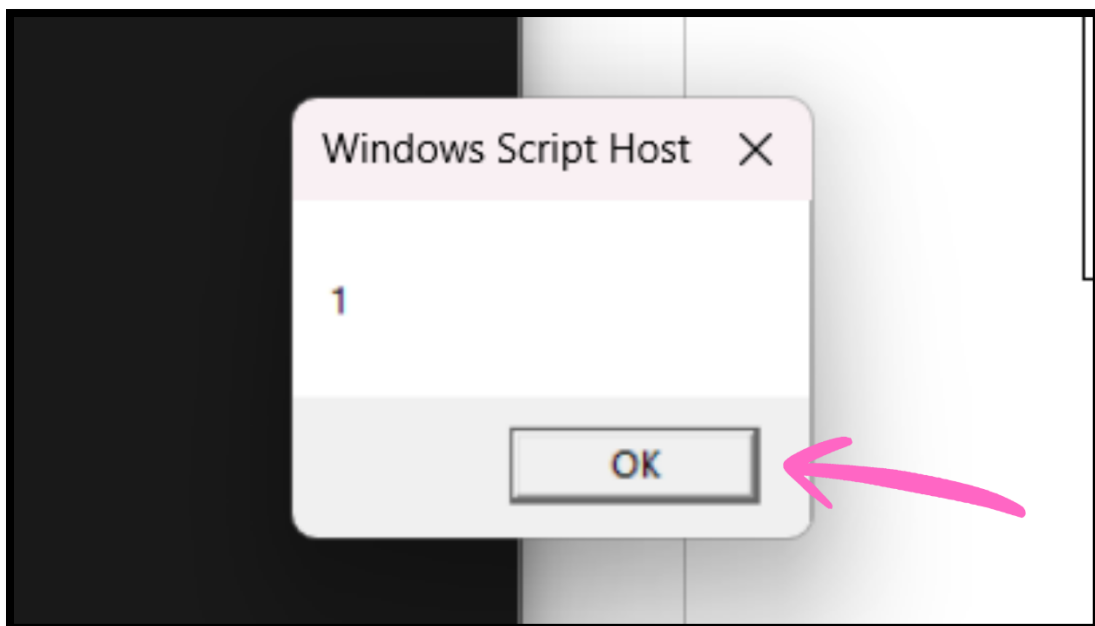




9. Double-click the *Test.vbs* file.

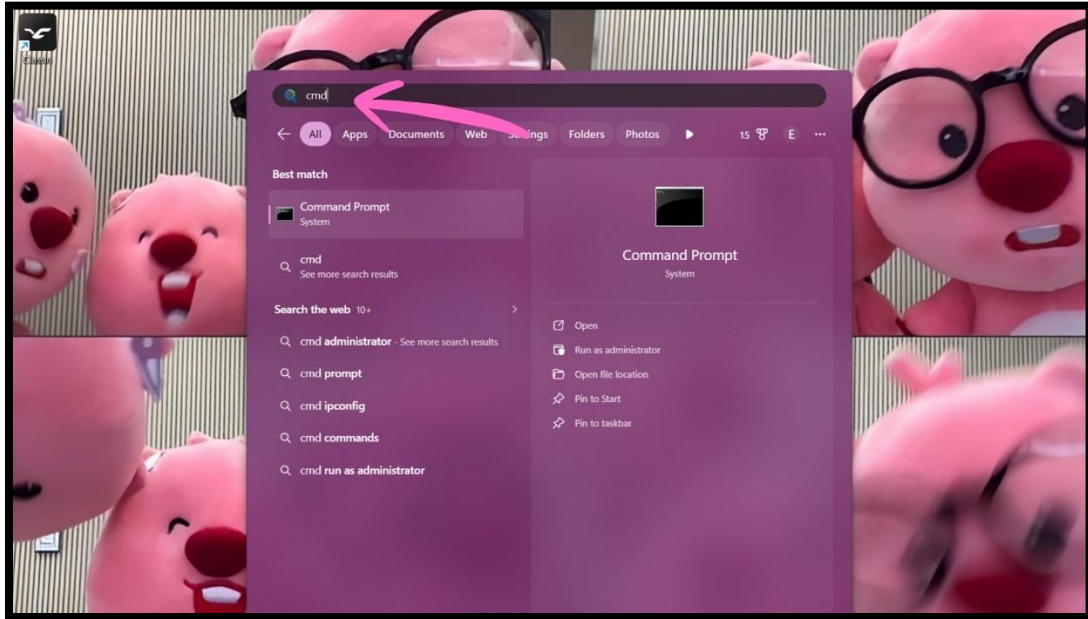


10. The script runs and a new window opens that displays the number 1. Click **OK**, and then number 2 displays in a new window. Continue until 5 has displayed.

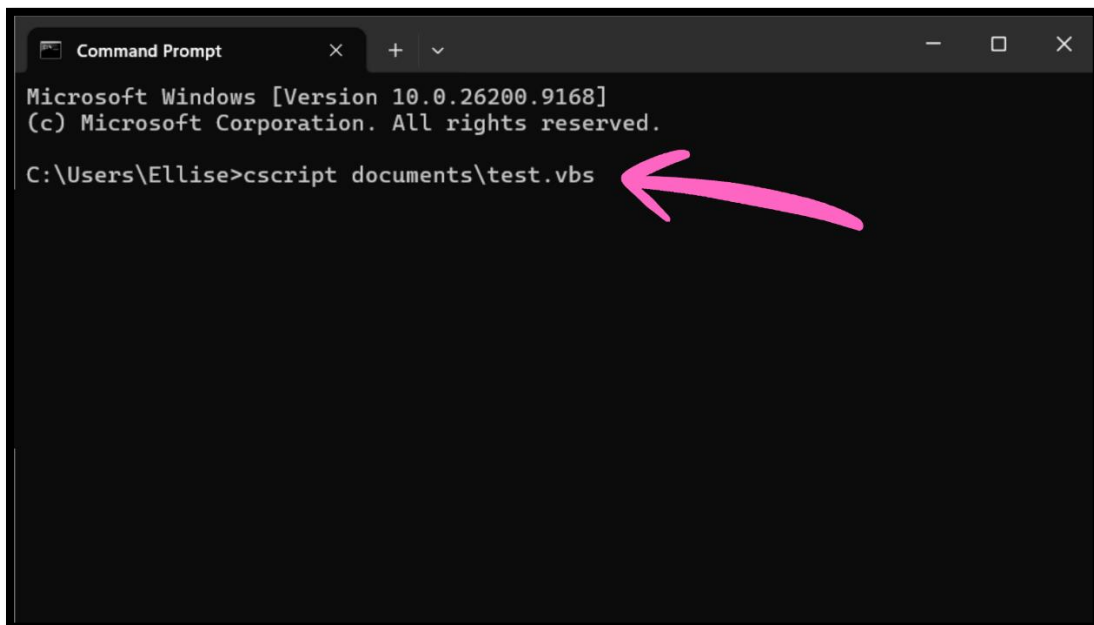




11. Click the **Start button**, type **cmd** in the search bar, and then click **Command Prompt** from the search results.



12. In the Command Prompt window, type *cscript documents\test.vbs* and then press ENTER.





13. The script runs from the command line and displays each number in sequence.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Ellise>cscript documents\test.vbs
Microsoft (R) Windows Script Host Version 10.0
Copyright (C) Microsoft Corporation. All rights reserved.

1
2
3
4
5

C:\Users\Ellise>
```

14. Switch to the Notepad window and update the script to read as shown below:

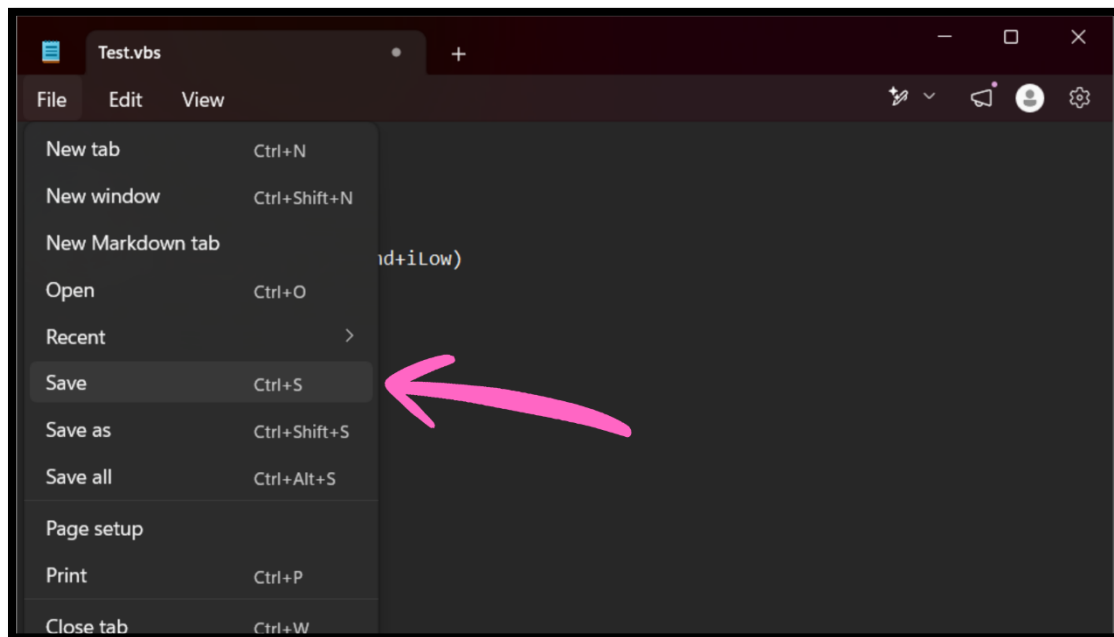
```
iHigh=50
iLow=10
for i=1 to 5
    randomize
    iDisplay=int((iHigh-iLow+1)*rnd+iLow)
    wscript.echo iDisplay
next
```

- This script uses some additional integer variables and built-in VBScript functions to return a random number between 10 and 50 during each iteration of the loop.

```
Test.vbs
File Edit View
iHigh=50
iLow=10
for i=1 to 10
    randomize
    iDisplay=int((iHigh-iLow+1)*rnd+iLow)
    wscript.echo iDisplay
next
```

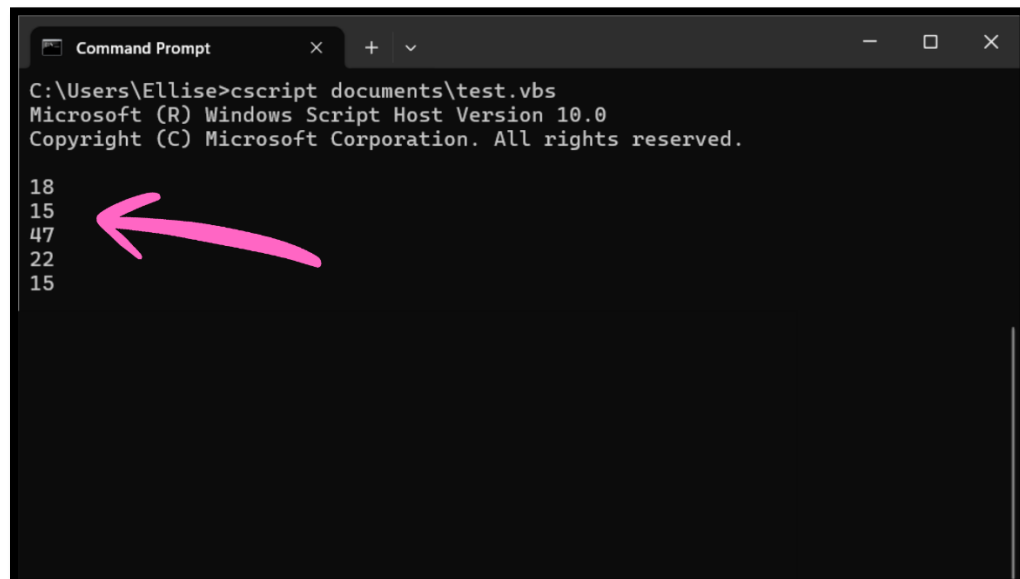


15. Save the file.



16. In the Command Prompt window, type `cscript documents\test.vbs` and then press ENTER.

- Five random numbers between 10 and 50 display.



17. What part of the code would you change to show 10 random numbers?

ANSWER: _____



18. Edit this line and save the file:

```
for i=1 to 10
```

```
Test.vbs
File Edit View
iHigh=50
iLow=10
for i=1 to 10
  randomize
  iDisplay=int((iHigh-iLow+1)*rnd+iLow)
  wscript.echo iDisplay
next
```

19. Run the script again.

```
Copyright (C) Microsoft Corporation. All rights reserved.

18
15
47
22
15

C:\Users\Ellise>cscript documents\test.vbs
Microsoft (R) Windows Script Host Version 10.0
Copyright (C) Microsoft Corporation. All rights reserved.

44
41
32
48
40
33
44
11
39
38
```



20. Switch to the Notepad window and update the script to read as shown below:

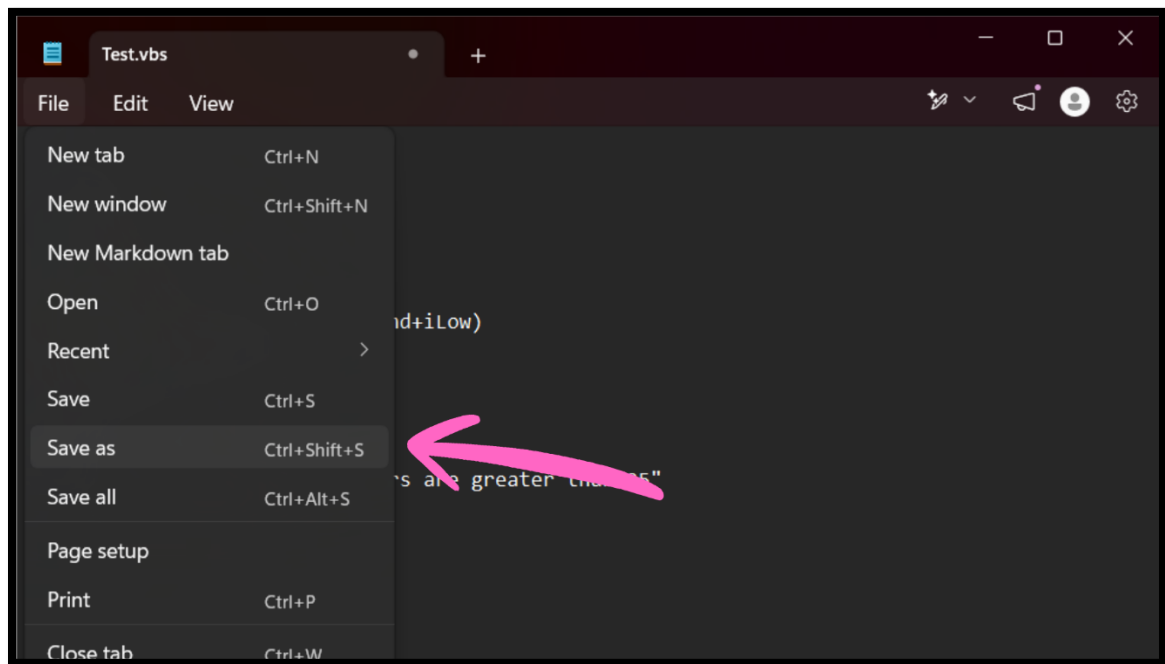
```
iHigh=50
iLow=10
iCount=0
sTitle = "Number count"
for i=1 to 10
    randomize
    iDisplay=int((iHigh-iLow+1)*rnd+iLow)
    wscript.echo iDisplay
    if iDisplay > 25 then
        iCount=iCount+1
    end if
next
sMsg = cstr(iCount) + " numbers are greater than 25"
msgbox sMsg, vbOK,sTitle
```

- These modifications add another branch to increment a variable if another variable exceeds a certain value. Note how the branching statements are nested. We also use some string-type variables and a function to convert an integer (*iCount*) into a string for use as text output in a message box. Finally, note the use of *vbOK* in the last line. This is an example of a constant. It defines the type of message box to display (one with just an OK button).

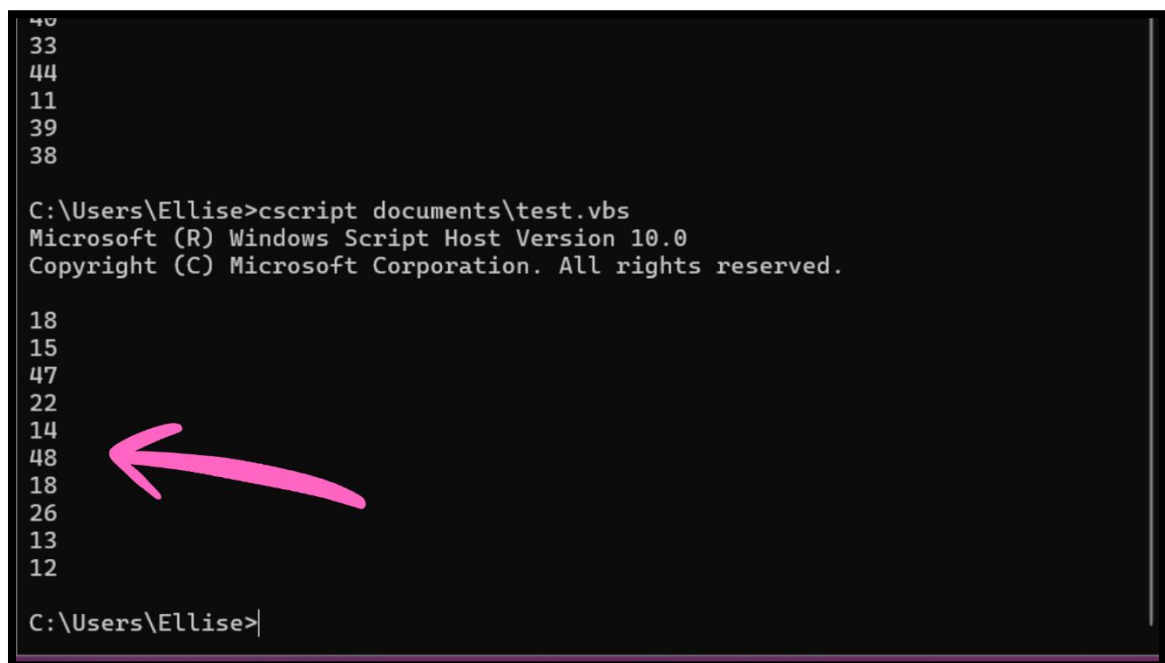
```
Test.vbs
File Edit View
iHigh=50
iLow=10
iCount=0
sTitle = "Number count"
for i=1 to 10
    randomize
    iDisplay=int((iHigh-iLow+1)*rnd+iLow)
    wscript.echo iDisplay
    if iDisplay > 25 then
        iCount=iCount+1
    end if
next
sMsg = cstr(iCount) + " numbers are greater than 25"
msgbox sMsg, vbOK,sTitle
```



21. Save the file.

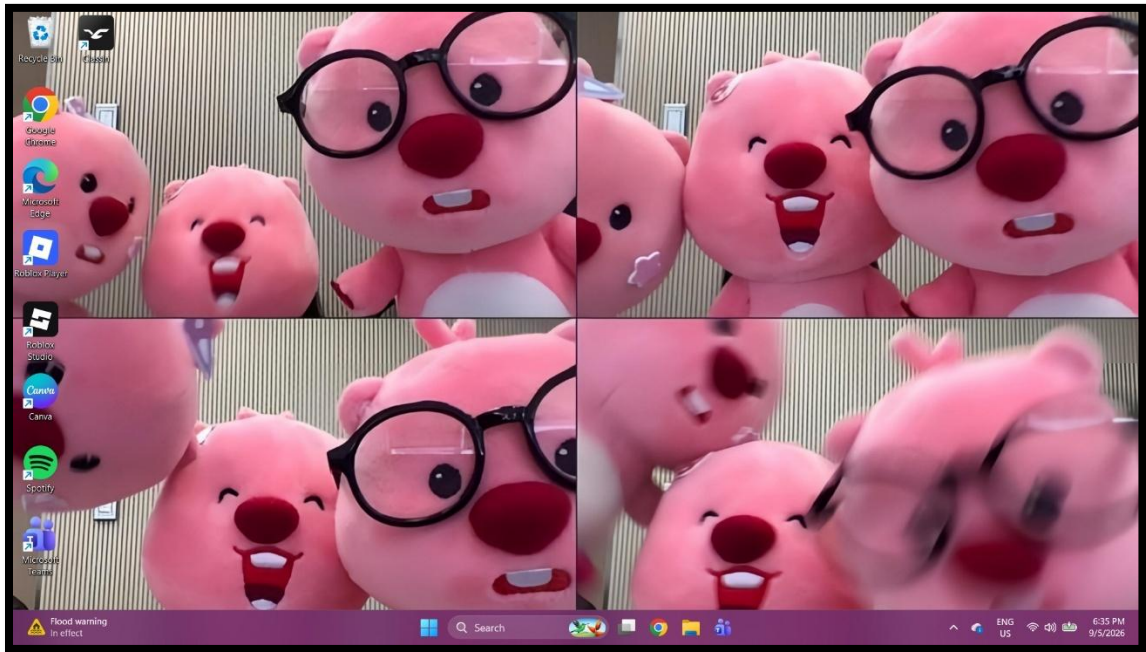


22. In the Command Prompt window, run the script.



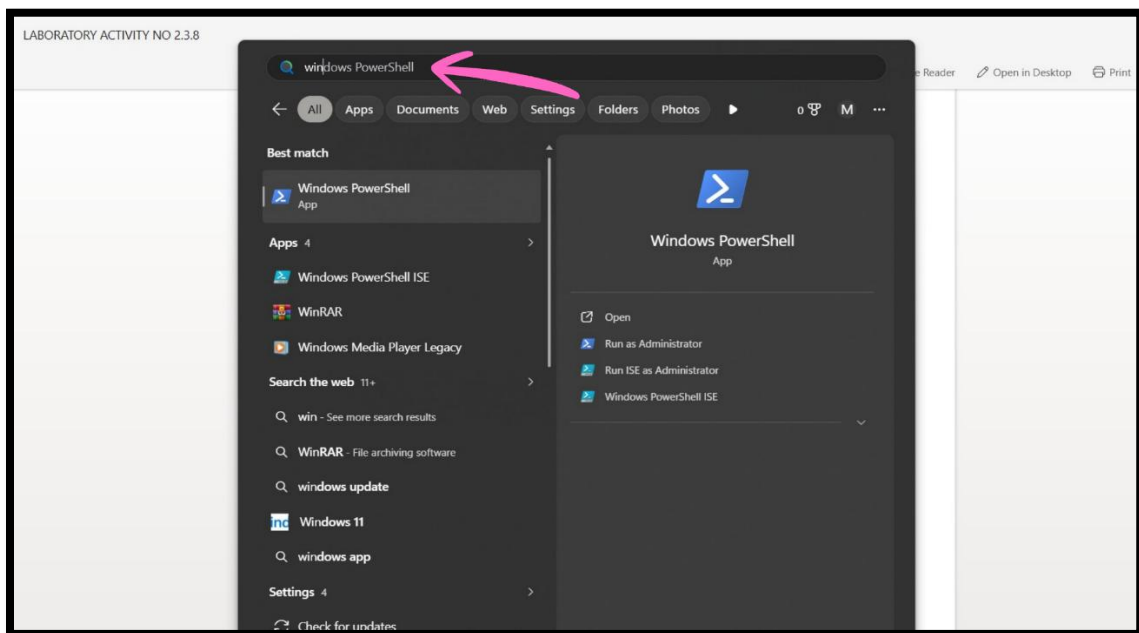


23. Close all open apps and windows.



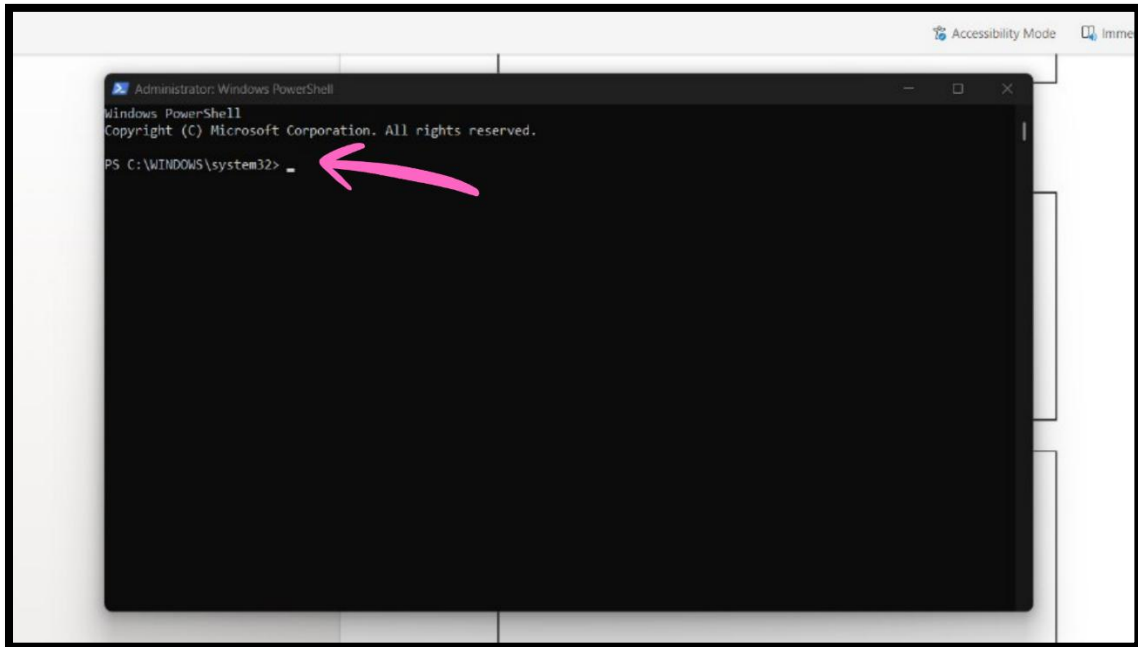
Part 3: Creating a Windows PowerShell Script

1. Click the **Start** button, type Windows PowerShell, right-click **Windows PowerShell** in the search results, and then click **Run as administrator**.



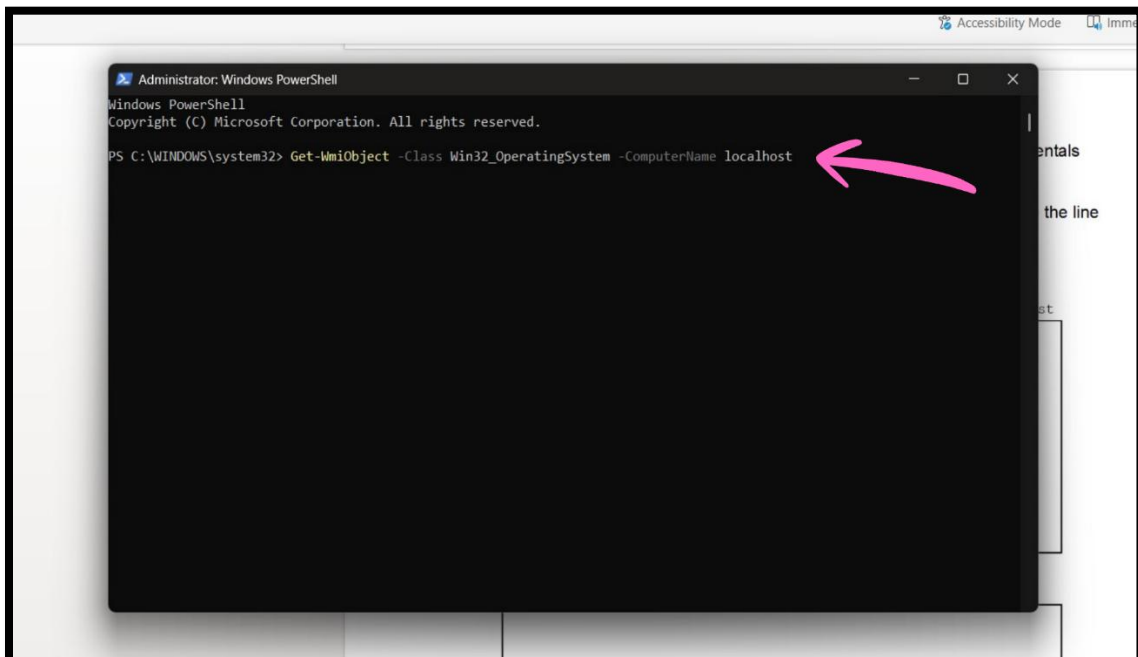


2. In the User Account Control window, click **Yes**.



3. In the Windows PowerShell window, type the following command (ignore the line break) and then press ENTER:

```
Get-WmiObject -Class Win32_OperatingSystem -ComputerName localhost
```





4. Examine the returned results.

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> Get-WmiObject -Class Win32_OperatingSystem -ComputerName localhost

SystemDirectory : C:\WINDOWS\system32
Organization     :
BuildNumber      : 26200
RegisteredUser   : Babs
SerialNumber     : 00342-42361-87882-AAOEM
Version          : 10.0.26200

PS C:\WINDOWS\system32>
```

5. In the Windows PowerShell window, type the following command and then press ENTER:

```
Get-WmiObject -Class Win32_OperatingSystem -ComputerName localhost |
Select-Object -Property CSName
```

- To save a bit of time, you can press the UP ARROW to select the original command and then edit it.

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> Get-WmiObject -Class Win32_OperatingSystem -ComputerName localhost

SystemDirectory : C:\WINDOWS\system32
Organization     :
BuildNumber      : 26200
RegisteredUser   : Babs
SerialNumber     : 00342-42361-87882-AAOEM
Version          : 10.0.26200

PS C:\WINDOWS\system32> Get-WmiObject -Class Win32_OperatingSystem -ComputerName localhost | Select-Object -Property CSName
```



6. Examine the returned results.
 - Your computer name is displayed in a table. The command uses a pipe (|) to send the output from one command to another command.

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> Get-WmiObject -Class Win32_OperatingSystem -ComputerName localhost

SystemDirectory : C:\WINDOWS\system32
Organization     : 
BuildNumber      : 26200
RegisteredUser   : Babs
SerialNumber     : 00342-42361-87882-AAOEM
Version          : 10.0.26200

PS C:\WINDOWS\system32> Get-WmiObject -Class Win32_OperatingSystem -ComputerName localhost | Select-Object -Property CSName
CSName
-----
DESKTOP-37KH317

PS C:\WINDOWS\system32> Get-Service
```

7. In the Windows PowerShell window, type the following command and then press ENTER:
Get-Service

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> Get-WmiObject -Class Win32_OperatingSystem -ComputerName localhost

SystemDirectory : C:\WINDOWS\system32
Organization     : 
BuildNumber      : 26200
RegisteredUser   : Babs
SerialNumber     : 00342-42361-87882-AAOEM
Version          : 10.0.26200

PS C:\WINDOWS\system32> Get-WmiObject -Class Win32_OperatingSystem -ComputerName localhost | Select-Object -Property CSName
CSName
-----
DESKTOP-37KH317

PS C:\WINDOWS\system32> Get-Service
```



8. Examine the returned results. A list of all services on your computer is displayed.

```
PS C:\WINDOWS\system32> Get-Service
```

Status	Name	DisplayName
Stopped	AarSvc_67f650bc	Agent Activation Runtime_67f650bc
Stopped	ADPSvc	Aggregated Data Platform Service
Stopped	ALG	Application Layer Gateway Service
Stopped	AppIDSvc	Application Identity
Running	AppInfo	Application Information
Stopped	AppReadiness	App Readiness
Running	AppXSvc	AppX Deployment Service (AppXSVC)
Stopped	ApXSvc	Windows Virtual Audio Device Proxy ...
Running	AudioEndpointBu...	Windows Audio Endpoint Builder
Running	AudioSrv	Windows Audio
Stopped	autotimesvc	Cellular Time
Stopped	AxInstSV	ActiveX Installer (AxInstSV)
Stopped	BcastDVRUserSer...	GameDVR and Broadcast User Service...
Running	BDESVC	BitLocker Drive Encryption Service
Running	BFE	Base Filtering Engine
Running	BITS	Background Intelligent Transfer Ser...
Running	BluetoothUserSe...	Bluetooth User Support Service_67f6...
Running	BrokerInfrastru...	Background Tasks Infrastructure Ser...
Running	BTAGService	Bluetooth Audio Gateway Service
Running	BthAvctpSvc	AVCTP service
Running	bthserv	Bluetooth Support Service
Running	camsvc	Capability Access Manager Service
Stopped	CaptureService_...	CaptureService_67f650bc
Running	cbdhsvc_67f650bc	Clipboard User Service_67f650bc
Running	CDPSvc	Connected Devices Platform Service

9. In the Windows PowerShell window, type the following command and then press ENTER:

```
Get-Service | Select DisplayName
```

```
PS C:\WINDOWS\system32> Get-Service | Select DisplayName
```

Status	DisplayName
Stopped	Windows Remote Management (WS-Manag...
Stopped	Windows Insider Service
Running	WLAN AutoConfig
Stopped	Microsoft Account Sign-in Assistant
Stopped	Local Profile Assistant Service
Stopped	Windows Management Service
Stopped	WMI Performance Adapter
Running	Intel(R) Management Engine WMI Prov...
Stopped	Windows Media Player Network Sharin...
Stopped	Work Folders
Stopped	Parental Controls
Stopped	Portable Device Enumerator Service
Running	Windows Push Notifications System S...
Running	Windows Push Notifications User Ser...
Running	WSAIFabricSvc
Running	Security Center
Running	Windows Search
Stopped	Windows Update
Running	Microsoft Usage and Quality Insights
Stopped	WWAN AutoConfig
Stopped	Xbox Live Auth Manager
Stopped	Xbox Live Game Save
Stopped	Xbox Accessory Management Service
Stopped	Xbox Live Networking Service



10. Examine the returned results. The same list of services is displayed but this time the output is filtered down to the service's "friendly" name.

```
Administrator: Windows PowerShell

PS C:\WINDOWS\system32> Get-Service | Select DisplayName

DisplayName
-----
Agent Activation Runtime_67f650bc
Aggregated Data Platform Service
Application Layer Gateway Service
Application Identity
Application Information
App Readiness
AppX Deployment Service (AppXSVC)
Windows Virtual Audio Device Proxy Service
Windows Audio Endpoint Builder
Windows Audio
Cellular Time
ActiveX Installer (AxInstSV)
GameDVR and Broadcast User Service_67f650bc
BitLocker Drive Encryption Service
Base Filtering Engine
Background Intelligent Transfer Service
Bluetooth User Support Service_67f650bc
Background Tasks Infrastructure Service
Bluetooth Audio Gateway Service
AVCTP service
Bluetooth Support Service
Capability Access Manager Service
CaptureService_67f650bc
Clipboard User Service_67f650bc
Connected Devices Platform Service
```

11. In the Windows PowerShell window, type the following command and then press ENTER:

```
Get-WmiObject win32_service | Select DisplayName, State | Sort State, DisplayName
```

```
Administrator: Windows PowerShell

Microsoft Defender Antivirus Service
WinHTTP Web Proxy Auto-Discovery Service
Windows Management Instrumentation
Windows Remote Management (WS-Management)
Windows Insider Service
WLAN AutoConfig
Microsoft Account Sign-in Assistant
Local Profile Assistant Service
Windows Management Service
WMI Performance Adapter
Intel(R) Management Engine WMI Provider Registration
Windows Media Player Network Sharing Service
Work Folders
Parental Controls
Portable Device Enumerator Service
Windows Push Notifications System Service
Windows Push Notifications User Service_67f650bc
WSATFabricSvc
Security Center
Windows Search
Windows Update
Microsoft Usage and Quality Insights
WLAN AutoConfig
Xbox Live Auth Manager
Xbox Live Game Save
Xbox Accessory Management Service
Xbox Live Networking Service

PS C:\WINDOWS\system32> Get-WmiObject win32_service | Select DisplayName, State | Sort State, DisplayName
```

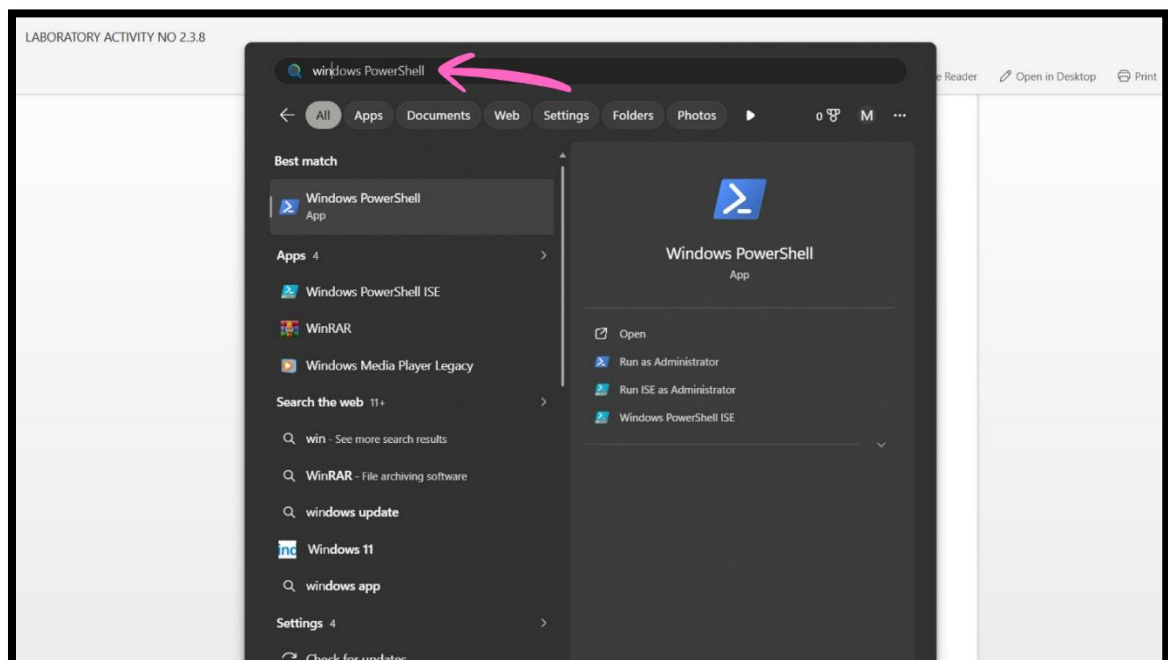


12. Examine the returned results. A list of all services on your computer is displayed, sorted by state and then display name.
- So far we have been running cmdlets individually. You can also create PowerShell scripts in the editor environment and save them as files.

```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> Get-WmiObject win32_service | Select DisplayName, State | Sort State, DisplayName

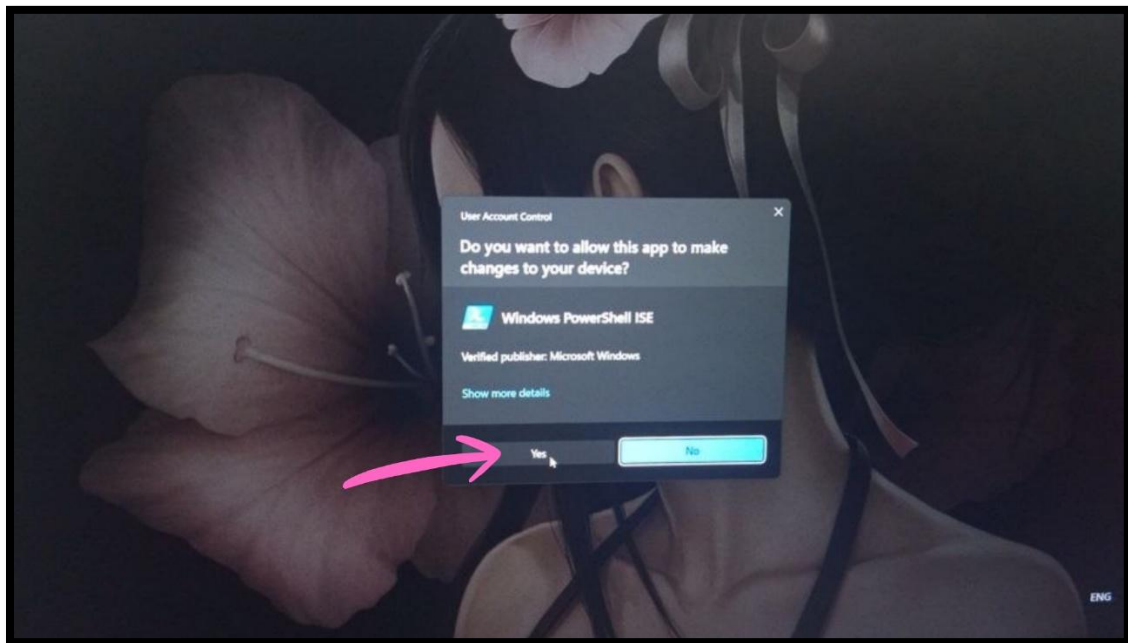
DisplayName                                     State
-----
Application Information                         Running
AVCTP service                                 Running
Background Intelligent Transfer Service         Running
Background Tasks Infrastructure Service          Running
Base Filtering Engine                          Running
BitLocker Drive Encryption Service              Running
Bluetooth Audio Gateway Service                Running
Bluetooth Support Service                      Running
Bluetooth User Support Service_67f650bc         Running
Capability Access Manager Service               Running
Client License Service (ClipSVC)               Running
Clipboard User Service_67f650bc                Running
CNG Key Isolation                             Running
COM+ Event System                             Running
Connected Devices Platform Service              Running
Connected Devices Platform User Service_67f650bc Running
Connected User Experiences and Telemetry        Running
CoreMessaging                                  Running
Credential Manager                             Running
Cryptographic Services                         Running
Data Sharing Service                           Running
Data Usage                                      Running
DCOM Server Process Launcher                    Running
Delivery Optimization                           Running
Dell Client Management Service                  Running
```

13. Click the **Start** button, type Windows PowerShell ISE, right-click **Windows PowerShell ISE** in the search results, and then click **Run as administrator**.





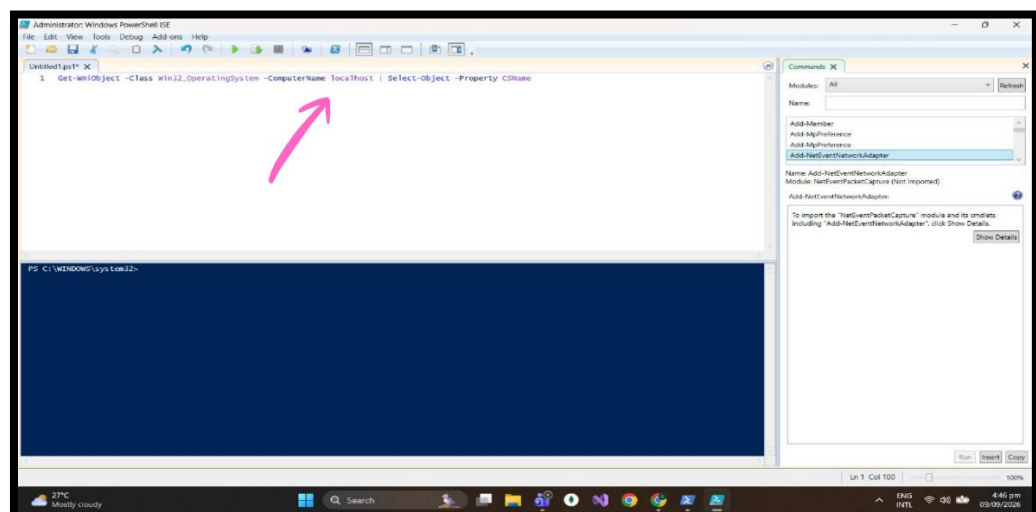
14. In the User Account Control window, click **Yes**.



15. In the Untitled1.ps1 area, type the following code as a single line:

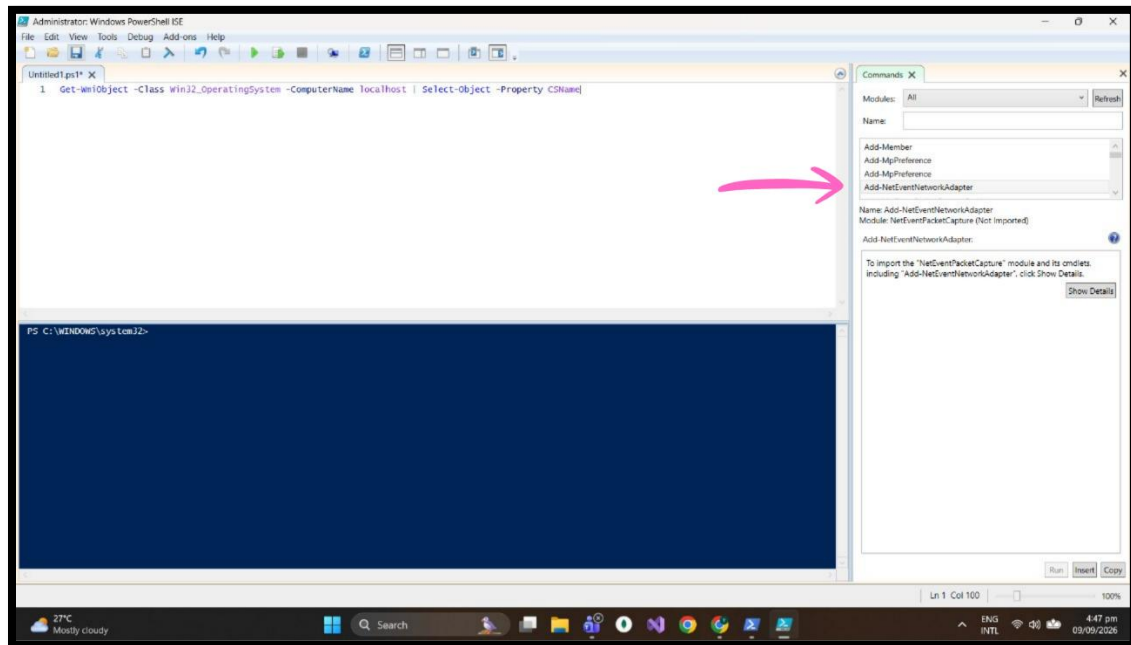
```
Get-WmiObject -Class Win32_OperatingSystem -ComputerName localhost |  
Select-Object -Property CSName
```

- Note that as you type, suggestion boxes appear helping you to autocomplete statements. Also notice that the editor applies color-coding to distinguish parts of your code. These features make a development environment much easier to work in than a plain text editor like Notepad.

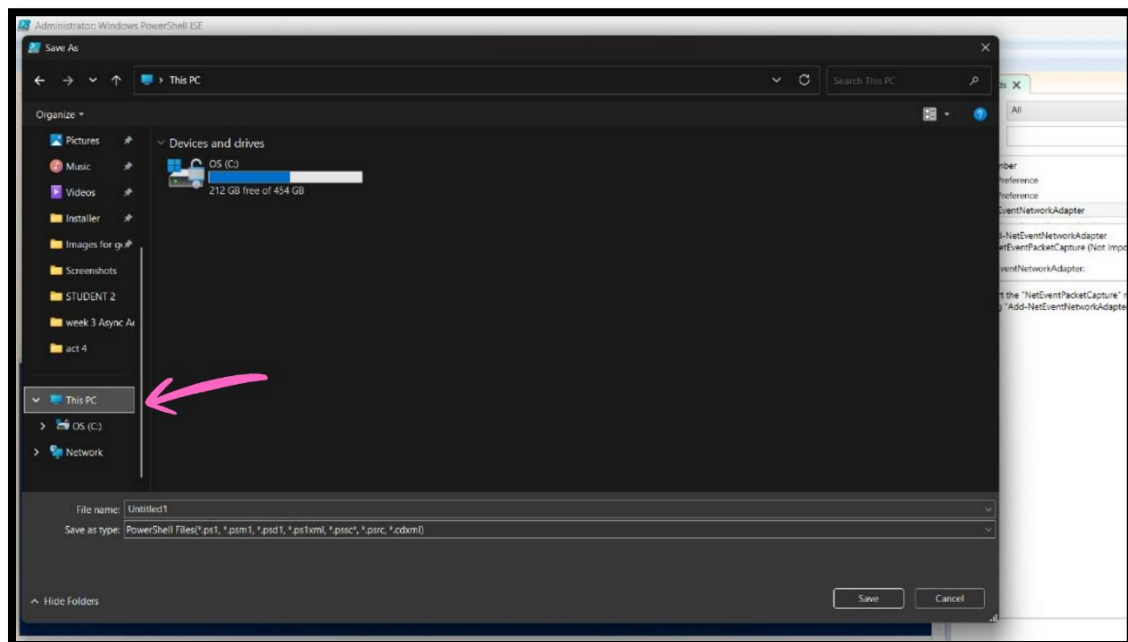




16. On the toolbar, click **Save**.

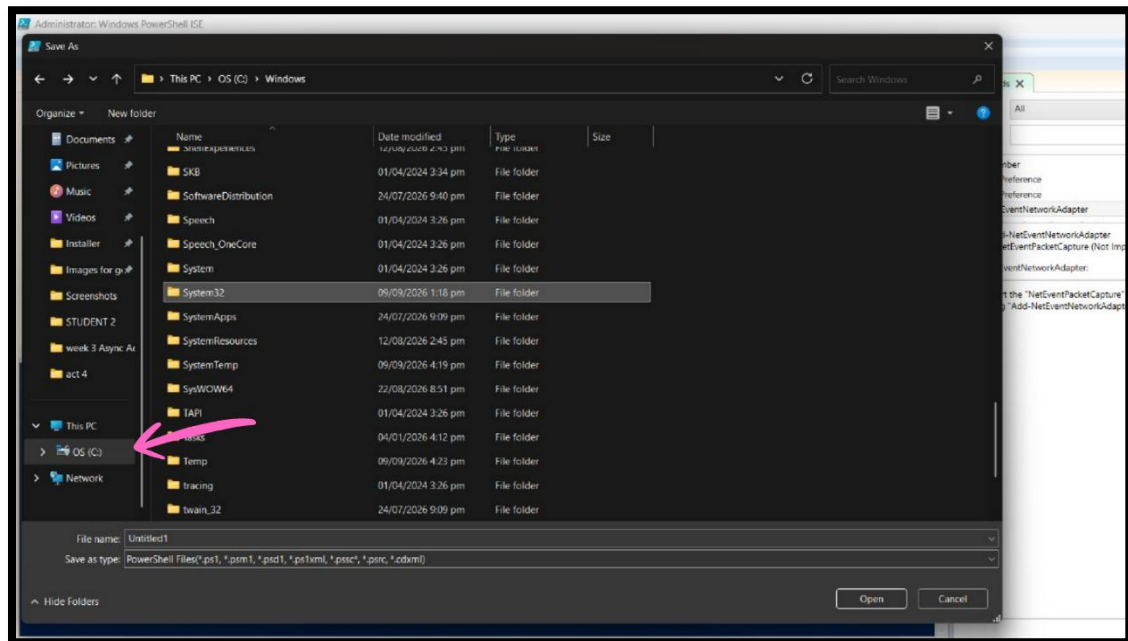


17. In the Save As window, in the navigation pane, expand **This PC**.

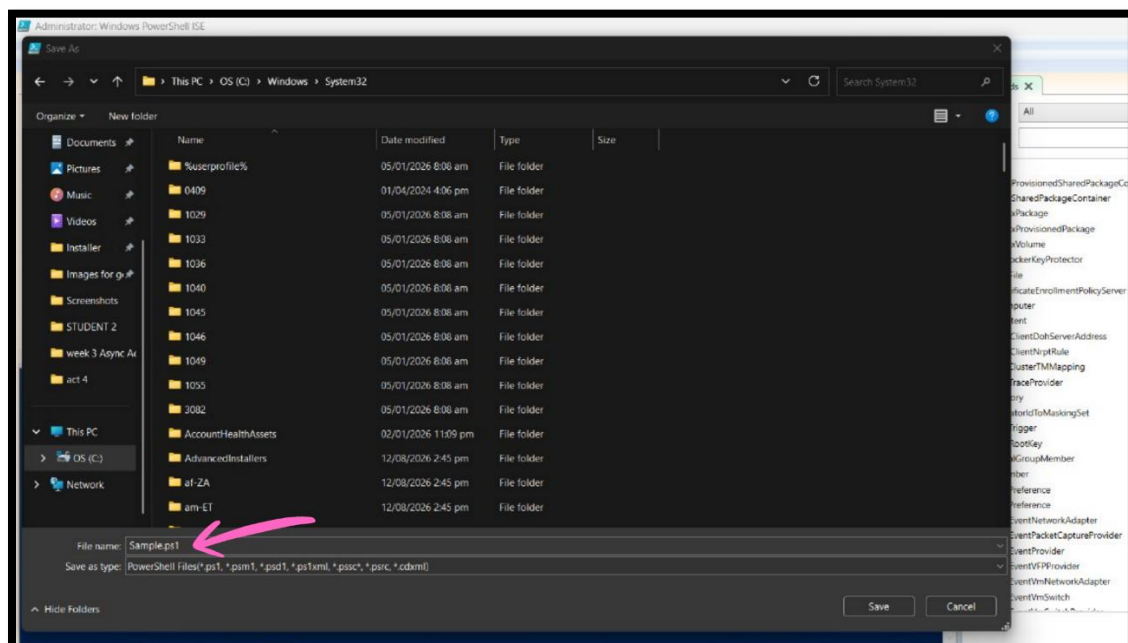




18. Expand **Local Disk (C:)**, expand **Windows**, and then click **System32**.

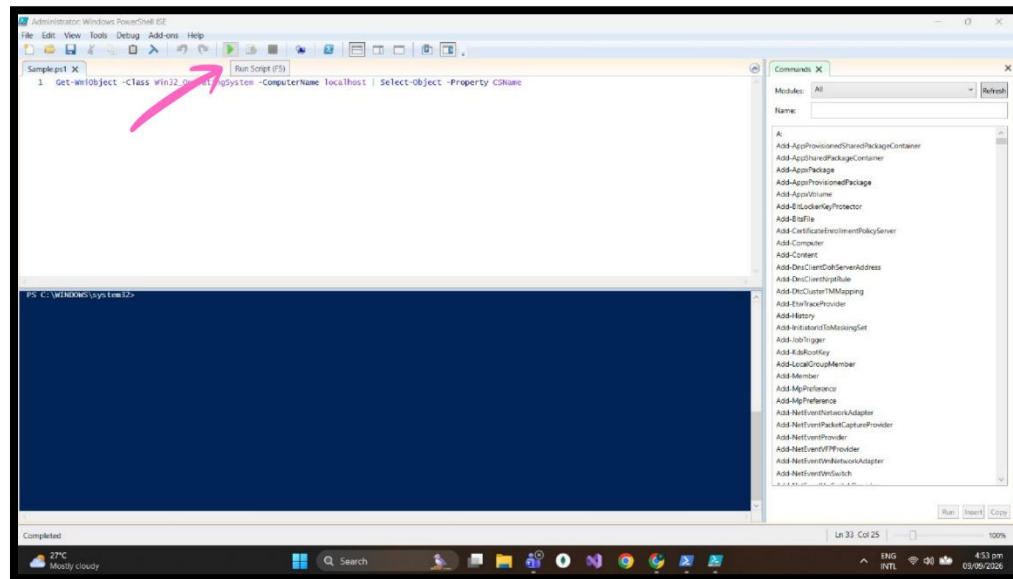


19. In the File name box, type **Sample.ps1** and then click **Save**.



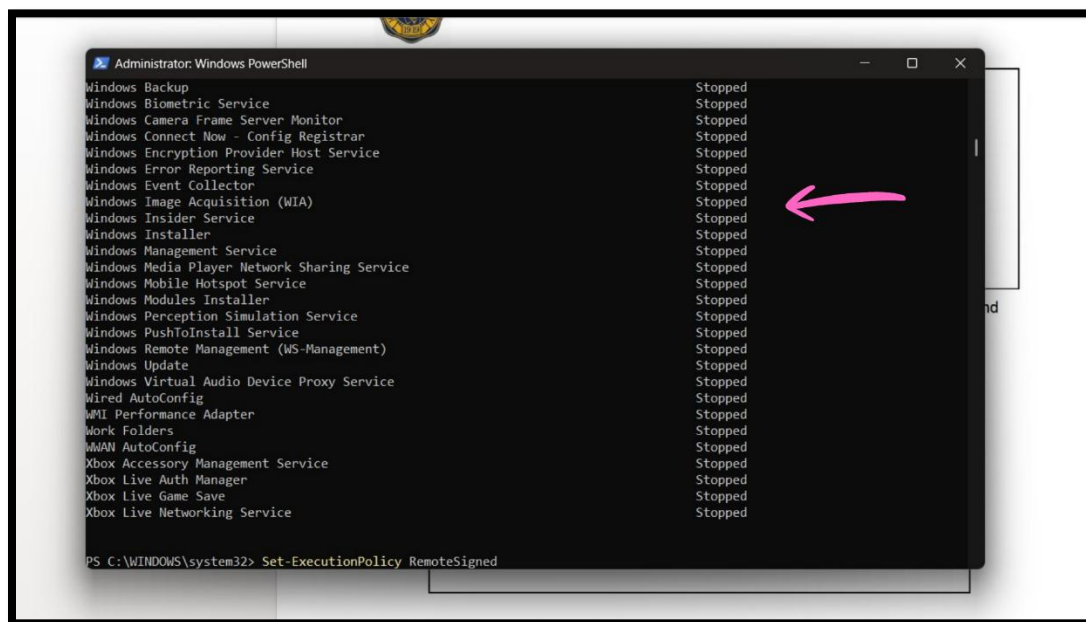


20. In the Administrator: Windows PowerShell ISE window, on the toolbar, click **Run**.
- You receive an error. This is because unsigned scripts are blocked by default. To enable scripts without a valid coding certificate to run, switch to the Windows PowerShell window.



21. In the Administrator: Windows PowerShell window, type the following command and then press ENTER:

Set-ExecutionPolicy RemoteSigned





22. When prompted, type A and press ENTER.

Administrator: Windows PowerShell

Service Name	Status
Windows Event Collector	Stopped
Windows Image Acquisition (WIA)	Stopped
Windows Insider Service	Stopped
Windows Installer	Stopped
Windows Management Service	Stopped
Windows Media Player Network Sharing Service	Stopped
Windows Mobile Hotspot Service	Stopped
Windows Modules Installer	Stopped
Windows Perception Simulation Service	Stopped
Windows PushToInstall Service	Stopped
Windows Remote Management (WS-Management)	Stopped
Windows Update	Stopped
Windows Virtual Audio Device Proxy Service	Stopped
Wired AutoConfig	Stopped
WMI Performance Adapter	Stopped
Work Folders	Stopped
WiMAX AutoConfig	Stopped
Xbox Accessory Management Service	Stopped
Xbox Live Auth Manager	Stopped
Xbox Live Game Save	Stopped
Xbox Live Networking Service	Stopped

PS C:\WINDOWS\system32> Set-ExecutionPolicy RemoteSigned

Execution Policy Change
The execution policy helps protect you from scripts that you do not trust. Changing the execution policy might expose you to the security risks described in the about_Execution_Policies help topic at <https://go.microsoft.com/fwlink/?LinkID=135170>. Do you want to change the execution policy?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"): A

23. Switch to Windows PowerShell ISE and click **Run**. Your simple one-line script runs.

Administrator: Windows PowerShell ISE

Sample.ps1 X

```
1. Get-WmiObject -Class Win32_OperatingSystem -ComputerName localhost | Select-Object -Property CSName
```

PS C:\WINDOWS\system32> C:\Windows\System32\Sample.ps1

CSName
DESKTOP-37KH317

PS C:\WINDOWS\system32>

Completed

Ln 41 Col 25 100%

27°C Mostly cloudy 4:55 pm 09/09/2026



24. Switch to the Windows PowerShell window.

```
Administrator: Windows PowerShell

Windows Image Acquisition (WIA)           Stopped
Windows Insider Service                   Stopped
Windows Installer                         Stopped
Windows Management Service                Stopped
Windows Media Player Network Sharing Service Stopped
Windows Mobile Hotspot Service            Stopped
Windows Modules Installer                 Stopped
Windows Perception Simulation Service      Stopped
Windows PushToInstall Service             Stopped
Windows Remote Management (WS-Management) Stopped
Windows Update                           Stopped
Windows Virtual Audio Device Proxy Service Stopped
Wired AutoConfig                          Stopped
WMI Performance Adapter                   Stopped
Work Folders                             Stopped
WMAN AutoConfig                           Stopped
Xbox Accessory Management Service          Stopped
Xbox Live Auth Manager                    Stopped
Xbox Live Game Save                       Stopped
Xbox Live Networking Service               Stopped

PS C:\WINDOWS\system32> Set-ExecutionPolicy RemoteSigned

Execution Policy Change
The execution policy helps protect you from scripts that you do not trust. Changing the execution policy might expose
you to the security risks described in the about_Execution_Policies help topic at
https://go.microsoft.com/fwlink/?LinkID=135170. Do you want to change the execution policy?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"): A
PS C:\WINDOWS\system32>
```

25. At the prompt, type Sample.ps1 and then press Enter. Your simple one-line script runs.

```
Administrator: Windows PowerShell

Windows Perception Simulation Service       Stopped
Windows PushToInstall Service              Stopped
Windows Remote Management (WS-Management) Stopped
Windows Update                            Stopped
Windows Virtual Audio Device Proxy Service Stopped
Wired AutoConfig                          Stopped
WMI Performance Adapter                   Stopped
Work Folders                             Stopped
WMAN AutoConfig                           Stopped
Xbox Accessory Management Service          Stopped
Xbox Live Auth Manager                    Stopped
Xbox Live Game Save                       Stopped
Xbox Live Networking Service               Stopped

PS C:\WINDOWS\system32> Set-ExecutionPolicy RemoteSigned

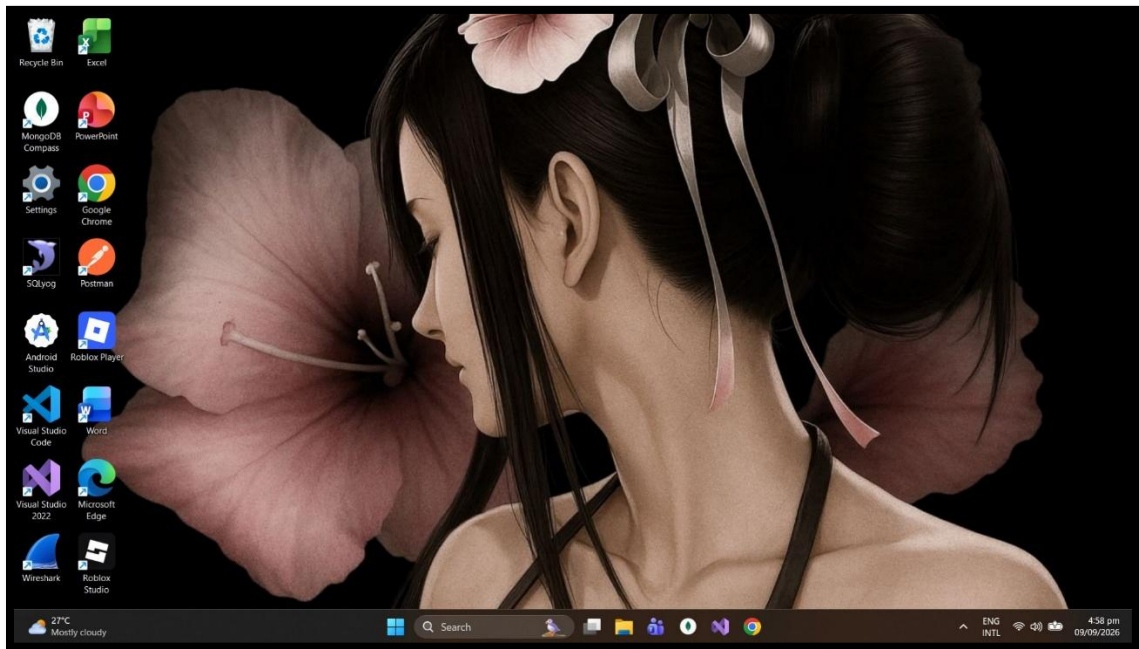
Execution Policy Change
The execution policy helps protect you from scripts that you do not trust. Changing the execution policy might expose
you to the security risks described in the about_Execution_Policies help topic at
https://go.microsoft.com/fwlink/?LinkID=135170. Do you want to change the execution policy?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"): A
PS C:\WINDOWS\system32> Sample.ps1

CSName
-----
DESKTOP-37KH317

PS C:\WINDOWS\system32>
```



26. Close all open windows and apps.





V. REFLECTION:

Upon successful completion of "Lab 2.3.8: Using Scripting Engine," write a comprehensive reflection (between 150-250 words) addressing the following points:

- **Learning Outcomes:** What new knowledge or skills did you acquire or reinforce regarding the use of a scripting engine?
- **Challenges and Solutions:** Describe any difficulties or unexpected issues you encountered during the lab. How did you overcome them, or what steps would you take to resolve them in the future?
- **Real-world Relevance:** How do the skills and understanding gained in this lab relate to real-world scenarios, either in your personal life or a professional IT context?
- **Critical Thinking:** If you had to use an internet image for any step, explain why you couldn't perform the action physically and how finding the image contributed to your understanding.

This activity was beneficial for me as I saw the practical application of scripting tools in computers. I gained more insight into the use of batch files, VBScript, and PowerShell. The activity also helped me learn loops, variables, conditions, and commands. It was initially hard to me to understand the commands and how to apply them practically. However, after following the instructions step by step, I successfully performed the activity. From this task, I discovered why scripting is critical to integrated information technology. It allows programmers to create codes that can execute repeatedly automatically instead of doing the same task manually. I think this activity is essential because it has helped me further understand scripting and its practical implications in the field.